Sequence-Based Machine Learning Models of Transcriptional Activator, Repressor and PADI-Score Prediction

Classification and regression modeling across plant and yeast activator, non-activator and repressor sequence datasets

Prepared by Anya Lachwani

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Summary of Work Accomplished (Below this summary is the detailed report)

Whether a protein sequence acts as a transcriptional activator, whether it represses, and how strongly it does either, cannot be read off the amino-acid string by inspection. Over six months this work built the machinery to learn that relationship computationally — and then applied it across five distinct biological questions. The output is not a single model: it is a reusable sequence-to-prediction pipeline, seven interchangeable feature representations, four purpose-built feature-analysis methods, a best-in-class PADI-score predictor, and a set of interpretive findings that converge on the same chemistry from independent directions.

Headline results

P = precision, R = recall

Task	Best model	Result
PADI-score prediction from sequence	Ridge regression on ESM protein-language-model embeddings	83.09% of predictions within ±0.5 and 58.35% within ±0.2 of the experimental score — the strongest result of the internship
Plant-specific vs. shared activators	Random Forest on the annotated feature set (top-100 selected)	81.935% accuracy; class 0 P 0.83 / R 0.85, class 1 P 0.80 / R 0.78 — both classes recalled evenly
Yeast-specific vs. shared activators	Random Forest on the annotated feature	79.89% accuracy; P 0.85 / R 0.88 on the majority class
Yeast-specific vs. plant-specific	Random Forest on the annotated feature	72.00% accuracy; yeast-specific activators identified at R 0.89
Plant activators vs. non-activators	Extra Trees regression on enrichment, then thresholding	Several implementations tried; working collaboration

1. Foundations

Datasets. Three preprocessing scripts split the original data into plant activators, shared (plant + yeast) activators and plant-specific activators, yeast specific. With the plant non-activator, repressor and non-repressor sets these define every comparison below. Non-activators and non-repressors are distinct sets, and the comparisons using them are distinct tasks.

Seven sequence representations. Six were developed for the classification work — amino-acid frequencies; dipeptide frequencies (direct, gapped, longest runs); whole-sequence physicochemical properties via BioPython (Kyte–Doolittle hydrophobicity, isoelectric point, GRAVY, aromaticity, instability index, molecular weight, helix/sheet/turn fractions); intrinsic disorder (residue-based disorder fraction plus mean/SD/max of metapredict per-residue scores); sliding-window physicochemical features (3-residue window, stride 1); and Simon's annotated feature set (900+ features, NaNs mean-imputed). A seventh, ESM protein-language-model embeddings, was introduced later for the PADI work.

Interpretation methods. Random Forest impurity importance, SHAP beeswarm and mean-absolute ranking, univariate R^2 screening, leave-one-feature-out ablation, model-free dipeptide enrichment analysis, and threshold sweeps converting regression output into class labels.

2. Plant-specific vs. shared activators — the primary comparison

Category 0 = plant-specific activators, Category 1 = activators shared between plant and yeast. Models were developed in three stages.

(a) **Initial model — Random Forest on sliding-window physicochemical features.** SHAP: isoelectric point in window 12–14 ranks highest overall, followed by aromaticity in window 36–38. A high isoelectric point pushes toward plant-specific; high local aromaticity (windows 20–22, 35–37) and high GRAVY (window 29–31) push toward shared.

(b) **Four parallel feature-representation models.** Four Random Forest classifiers trained on the same sequences and labels, differing only in representation, so the contribution of the representation itself is isolated.

Random Forest	Feature representation	SHAP attribution recorded
Model 1	All features combined	isoelectric point, freq_D, aromaticity, freq_F, mean predicted disorder; frequencies D, F, W, L, K, R throughout and all three disorder summaries in the top 20
Model 2	Physicochemical properties only	isoelectric point (≈2× the next feature), aromaticity, disorder_mean, gravy, sheet_frac
Model 3	Amino-acid composition + dipeptides	Frequencies: freq_D, freq_F, then freq_L, freq_W (K and R oppose). Dipeptides: gapped D–L, longest W run, LD, DF, FD
Model 4	Integrated (physicochemical + frequencies + dipeptides + composition + disorder)	Annotated Features

Two supporting analyses ran alongside these: a model-free dipeptide enrichment analysis (frequency in shared minus plant-only, with a 20×20 heatmap) ranking DL, DD, DF, LD, FD highest in the shared set and QQ, SS, NN, GG, SG in the plant-specific set; and unsupervised k-means with PCA on the scaled physicochemical features, showing the classes overlapping in PC1–PC2 but forming partly distinct regions.

(c) **Model 4 — Random Forest on Simon's annotated feature set.** The final architecture for this comparison, after testing several modelling and feature-selection approaches: mean imputation of missing values → initial Random Forest on the complete annotated feature set → ranking by that model's feature importances → reduction to the top 100 features → final Random Forest on those 100 → evaluation on held-out test data → SHAP analysis of the final model.

Accuracy 81.935%. Confusion matrix [[301, 52], [60, 207]]; class 0 P 0.83 / R 0.85 / F1 0.84, class 1 P 0.80 / R 0.78 / F1 0.79 (353 plant-specific and 267 shared sequences in the test set). This is the best-separated of the three activator-group comparisons and the only one in which both classes are recalled comparably well.

3. The same pipeline applied to the other two activator-group comparisons

Comparison	Accuracy	Confusion matrix	Classification report
Yeast-specific vs. plant-specific	72.00%	[[116, 204], [74, 599]]	Class 0 (plant-specific): P 0.61 / R 0.36 / F1 0.45 (n=320); Class 1 (yeast-specific): P 0.75 / R 0.89 / F1 0.81 (n=673)
Yeast-specific vs. shared	79.89%	[[590, 83], [106, 161]]	Class 0 (yeast-specific): P 0.85 / R 0.88 / F1 0.86 (n=673); Class 1 (shared): P 0.66 / R 0.60 / F1 0.63 (n=267)

SHAP on the yeast-specific vs. plant-specific model is led by amino-acid entropy and aromatic fraction. In both comparisons the majority class is recalled well and the minority class poorly — the class-imbalance behaviour that became the central obstacle in the next workstream.

4. Plant + yeast activators vs. plant non-activators

Three Random Forest classifiers trained in parallel on the same sequences, each on a different representation and each interpreted through SHAP. The purpose was to compare which representation carried the discriminative signal.

Random Forest model	Representation	SHAP attribution recorded
Frequency model	Amino-acid frequencies only	freq_D dominates, then arginine, glutamate, lysine and tryptophan
Composition-class model	Five residue composition classes	Negative and positive classes carry almost all the signal and act in opposition; special, hydrophobic and polar contribute little
Dipeptide model	Direct + gapped dipeptides, longest	The DD pair, the longest aspartate run and the longest arginine run lead

All three encodings converge on the same chemistry from three directions: charge.

5. Plant activators vs. plant non-activators — a sequential investigation

Class 1 = plant activators, class 0 = plant non-activators. Each step was taken in response to the result of the previous one, so these are stages of one investigation rather than alternatives to be compared.

An initial Random Forest on annotated features returned under 20% recall on the activator class against over 99% recall on non-activators — a strong majority-class bias. Predicted activators were often correct, but most true activators were missed. Every subsequent step targets activator recall.

Imbalance mitigation, in order: class weighting at 2:1 then 3:1 toward the minority class (precision and recall traded off, no substantial gain); restricting training to the top 75% of activators (little change); Balanced Random Forest and GridSearchCV hyperparameter optimisation, repeated on the top-75% subset (remained ≈50%, matching several classifiers tested in Lucas's notebook); and a fine-tuned HistGradientBoosting classifier (performance similar to the Random Forest approaches).

Because the classifier was not the constraint, attention moved to the 900+ feature space: a SHAP-based reduction pipeline (train on all features → SHAP → rank → select → retrain → re-evaluate) was run at the top 100, 200 and 300 features. This produced a small improvement only, indicating that dimensionality alone did not explain the difficulty.

The problem was then reframed as regression following discussions with Hector's group: predict the continuous enrichment value first, then apply the activator boundary afterwards. Following Professor Max Staller's suggestion, a univariate Linear/Ridge regression was fitted for every annotated feature individually and features ranked by their own R^2; the strongest were charge_density, frac_pos, frac_R, XXcls_POS-HYDRO, XXcls_HYDRO-POS and XXcls_POS-NONPOL. The top 100 fed a multivariate Ridge regression predicting enrichment — Train R^2 0.1403, Test R^2 0.1379, MAE 0.4673, RMSE 0.6716, Spearman 0.4015 — whose output was classified at thresholds of 0.5, 1.0, 1.5, 2.0 and 2.5 rather than at one fixed boundary.

A leave-one-feature-out pipeline was then built as an alternative importance measure (ablate each feature, retrain, rank by the change in accuracy/precision/recall, keep the top 100, predict enrichment, apply manual thresholds). Following a second discussion with Juan and Jacob, **Lasso regression** was added for L1 sparse selection with SHAP run directly on the regression, and an **Extra Trees regressor** was trained to capture nonlinear relationships and interactions between annotated features and enrichment, giving a direct comparison between linear regularised models and tree-based ensemble regression but the classification recall and precision rates were not very good. Because the right way to proceed was not clear, the model was handed to Juan and Jacob rather than developed further.

6. Plant repressors vs. non-repressors — a separate feature-discovery task

This workstream addresses repression, and its design goal was different: following discussion with Simon, and given the imbalance of the dataset, the aim was to find which features contribute consistently rather than to maximise the predictive power of any single model. Multiple class-balanced batches were constructed — the repressor set held constant, a fresh sample of non-repressors drawn for each batch, equal numbers in each — and a separate Random Forest trained per batch, so that feature importance could be judged by its stability across independently trained models.

Three conditions were designed: (1) original sequences, as a baseline; (2) sequences with known motifs randomly mutated, to test whether the models depend on known motif signal; (3) sequences containing known motifs removed entirely, a stricter test of signal beyond the predefined motifs. Condition 3 is in progress.

7. PADI-score regression from protein sequence

Task: protein sequence → representation → regression model → predicted PADI score. An ensemble averaging a Random Forest Regressor and a CatBoost Regressor was built first on engineered physicochemical + positional-encoding features; ESM embeddings were then substituted into the same ensemble to test transfer learning.

Ensemble (Random Forest Regressor + CatBoost Regressor)	RMSE	MAE	R^2
Engineered physicochemical + positional features	0.772	0.429	0.389
ESM embeddings	0.784	0.437	0.371

Because the representation was not the binding constraint for that model family, the regressor was changed instead. A Ridge regression on the ESM embeddings — considerably simpler than the nonlinear ensemble and neural-network architectures — produced the strongest results of any model tested:

Model	Within ±0.2	Within ±0.3	Within ±0.4	Within ±0.5
Ridge Regression on ESM embeddings — best model	58.35%	72.26%	79.11%	83.09%

Infrastructure left behind

Every method above was built from scratch and is reusable beyond the question it was written for: three dataset-construction scripts; seven interchangeable sequence representations, including engineered metapredict disorder summaries and ESM embedding extraction; a SHAP-based feature-selection pipeline; a univariate R^2 screening pipeline; a leave-one-feature-out ablation pipeline; a batch-balanced sampling design for stable feature discovery; and a threshold-sweep procedure for turning any regression output into a tunable classifier. All models can be accessed through my GitHub. Please let me know if you would like access.

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1. Introduction

Transcription factors regulate gene expression through short protein regions that either switch target genes on (activation domains) or shut them down (repression domains). Which sequences carry that activity, and what makes one sequence a stronger regulator than another, is not readable directly from the amino-acid string by inspection. The work described in this report asks whether that relationship can be learned computationally: given a protein sequence and nothing else, can a machine-learning model predict whether it acts as an activator, whether it acts as a repressor, and how strong that activity is?

The datasets available for this work were experimentally labelled plant and yeast sequence sets. Sequences were labelled as activators or non-activators, as repressors or non-repressors, and for a subset with two continuous measures of activity: an enrichment value and a PADI score. Because some sequences appear as activators in both the plant and the yeast dataset while others are activator-labelled in only one organism, the data also allow a second, comparative question: are organism-specific activators chemically distinguishable from activators whose activity is shared across plant and yeast?

The work therefore had two related objectives, pursued in parallel throughout:

- Prediction — build models that assign an activator, repressor or activity-score label to an unseen protein sequence, and measure how accurate those assignments are.
- Interpretation — identify which sequence-derived features drive those predictions, so that the models produce biological hypotheses about what distinguishes one class from another rather than only a label.

The second objective is the reason so much of the work below concerns feature engineering and feature attribution rather than accuracy alone. In several of the comparisons the classifier was deliberately not optimized for maximum predictive performance, because the question being asked of it was "which sequence properties separate these two groups?" rather than "how well can this group membership be predicted?". Where that trade-off was made deliberately, it is stated in the relevant section.

1.1 Approach

All experiments share a common shape. Protein sequences are converted into a fixed-length numerical feature vector; a model is trained on those vectors against the experimental label; performance is measured on held-out data; and a feature-attribution method is applied to the trained model to rank which features mattered. The variation between experiments lies in three places: how the sequence is represented, which model is fitted, and whether the target is a discrete class or a continuous score.

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Protein sequence → Feature representation → Model → Prediction → Feature attribution
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Six sequence representations were developed and compared over the course of the classification work: amino-acid frequencies; dipeptide (adjacent amino-acid pair) frequencies; physicochemical properties; intrinsic-disorder features; sliding-window features; and a pre-existing annotated feature set of over 900 features supplied within the group. A seventh representation, embeddings from the pretrained ESM protein language model, was introduced later for the PADI-score work in Section 9. Model families spanned Random Forest and its balanced variant, HistGradientBoosting, Extra Trees, CatBoost, linear and regularised regression (Ridge, Lasso), a multi-layer perceptron and a convolutional network.

2. Keywords and Methods Index

The following terms describe the methods, features and metrics used in this work. They are provided as an index for readers scanning for a specific technique; each is defined in the glossary in Section 10.

Biological domain — Transcription factor; activation domain; repression domain; activator; non-activator; repressor; non-repressor; transcriptional enrichment value; PADI score; sequence motif; motif mutagenesis; motif ablation; plant–yeast conservation; organism-specific activity; amino acid; dipeptide; peptide.

Sequence features — Amino-acid composition; amino-acid frequency; dipeptide frequency; gapped dipeptide; longest homopolymeric run; sliding-window features; window size; stride; physicochemical properties; hydrophobicity; Kyte–Doolittle scale; GRAVY; isoelectric point (pI); charge density; fraction positive; fraction negative; fraction aromatic; fraction nonpolar; fraction hydrophobic; aromaticity; aromatic clustering; amino-acid entropy; instability index; molecular weight; helix fraction; sheet fraction; turn fraction; secondary-structure propensity; intrinsic disorder; disorder fraction; disorder-promoting residues; mean / standard deviation / maximum predicted disorder; residue-class pair features.

Tools and libraries — BioPython; metapredict; ESM protein language model; SHAP; CatBoost; GridSearchCV.

Models — Random Forest classifier; Balanced Random Forest; HistGradientBoosting; Extra Trees classifier and regressor; Random Forest regressor; CatBoost regressor; ensemble averaging; Linear regression; Ridge regression (L2); Lasso regression (L1); multi-layer perceptron (MLP); convolutional neural network (CNN); univariate regression; multivariate regression.

Training and model selection — Supervised learning; unsupervised learning; k-means clustering; principal component analysis (PCA); dimensionality reduction; transfer learning; pretrained embeddings; train–test split; held-out test set; cross-validation; hyperparameter tuning; GridSearchCV; regularisation; class imbalance; class weighting; balanced sampling; minority class; majority-class bias; decision-threshold optimisation; mean imputation; feature scaling; feature selection; univariate feature screening; feature ablation; leave-one-feature-out; batch-based balanced sampling; feature-importance stability; overfitting.

Interpretation — Feature importance; Gini / impurity-based importance; SHAP (SHapley Additive exPlanations); SHAP beeswarm plot; mean absolute SHAP; feature attribution; enrichment heatmap; distribution comparison; kernel density estimate.

Evaluation metrics — Accuracy; balanced accuracy; precision; recall; F1-score; support; confusion matrix; classification report; coefficient of determination (R^2); mean absolute error (MAE); root mean squared error (RMSE); Spearman rank correlation; error-tolerance band.

3. Executive Summary

Five workstreams were completed or advanced during the period. Each is summarized below with its headline outcome; full detail follows in Sections 5 to 9.

3.1 Activator-group classification (Section 5 and Section 6)

Random Forest classifiers were built and trained to test whether sequence differences separate the following pairs of activator groups:

- Plant-specific activators vs. activators shared between plant and yeast
- Plant-specific activators vs. yeast-specific activators
- Yeast-specific activators vs. shared plant + yeast activators
- Plant + yeast activators vs. plant non-activators

Six feature representations were developed and compared: amino-acid frequencies, dipeptide frequencies, physicochemical properties, disorder features, sliding-window features, and the annotated feature set. Random Forest feature importance and SHAP analysis were used to determine which features differentiated each group. The three groups tested against each other were separable to differing degrees, with test accuracies of 81.93% (plant-specific vs. shared, annotated features), 72.00% (yeast-specific vs. plant-specific, annotated features) and 79.89% (yeast-specific vs. shared, annotated features).

3.2 Plant activators vs. plant non-activators (Section 7)

Random Forest, Balanced Random Forest and HistGradientBoosting classifiers were built and compared. Class imbalance was investigated through class weighting, balanced sampling, threshold optimization, GridSearchCV and feature selection. When classification performance remained limited, the problem was reframed as enrichment-value regression followed by threshold-based classification, using Linear, Ridge, Lasso and Extra Trees models. SHAP, R^2-based feature ranking and feature ablation were used to identify important predictors.

3.3 Plant + yeast activators vs. plant non-activators (Section 8)

Three Random Forest models were built, each trained on a different feature representation of the same sequences — amino-acid frequencies, amino-acid composition classes, and dipeptides — and each was interpreted through SHAP to compare which representation carried the discriminative signal.

3.4 Plant repressors vs. non-repressors (Section 8.2)

Repeated Random Forest models were built on class-balanced batches to identify features that consistently differentiated repressors from non-repressors. Feature importance was compared across original sequences, sequences with known motifs randomly mutated, and sequences containing known motifs removed. The focus here was on robust feature discovery rather than maximizing classification accuracy.

3.5 PADI-score prediction (Section 9)

Regression models were developed to predict continuous PADI scores directly from protein sequence. Random Forest, CatBoost, Extra Trees, Ridge, MLP and CNN regressors were built and compared, and manually engineered sequence features were compared against ESM protein-language-model embeddings.

Best model to date: ESM embeddings combined with Ridge regression, placing 83.09% of predictions within ±0.5 and 58.35% within ±0.2 of the experimental PADI score.

4. Common Methodology

4.1 Dataset construction

Three preprocessing scripts were written to split the original data into the biologically relevant subsets used throughout:

1. Plant activators — the plant dataset filtered to retain only sequences labelled as activators.
2. Shared activators — activator sequences occurring in both the plant and the yeast dataset.
3. Plant-specific activators — plant activators that did not overlap with the yeast activator dataset.

These subsets, together with the plant non-activator, repressor and non-repressor sets, form the class definitions used in every comparison below.

4.2 Sequence feature engineering

Physicochemical properties

A feature-extraction pipeline was written to convert each protein sequence into a numerical vector of physicochemical descriptors, computed with BioPython. Features included hydrophobicity on the Kyte–Doolittle scale, isoelectric point, GRAVY (grand average of hydropathy), aromaticity, instability index, molecular weight and secondary-structure propensities (helix, sheet and turn fractions).

Sliding-window physicochemical features

To capture local rather than whole-sequence character, the same physicochemical properties were computed over a sliding window: a window of three amino acids was moved across each sequence with a stride of one position (positions 1–3, 2–4, 3–5, 4–6, and so on), and properties were calculated for each window. The resulting window-level measurements were concatenated into a numerical representation of the sequence. The classifier therefore never sees the raw amino-acid string; it sees a vector describing local physicochemical properties along the sequence.

Amino-acid composition and dipeptides

A second representation captured composition and short-range patterning: individual amino-acid frequencies, dipeptide (adjacent pair) frequencies, and frequencies of selected amino acids and sequence patterns.

Intrinsic disorder

Two disorder representations were developed. The first is a residue-based sequence-level disorder fraction, computed by identifying disorder-promoting residues and taking their frequency relative to sequence length:

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Disorder fraction = number of disorder-promoting residues / sequence length
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The second uses metapredict. Because metapredict returns one disorder score per position — N scores for a sequence of length N — its output is not directly usable as a fixed-length feature. Three summary features were therefore engineered from the per-position predictions: mean predicted disorder, standard deviation of predicted disorder, and maximum predicted disorder. Of these, mean predicted disorder showed the strongest predictive contribution in the initial feature-importance analysis, suggesting that the

overall degree of predicted disorder may be particularly informative for separating plant-specific from shared activators.

Integrated representation

A combined pipeline concatenated physicochemical properties, amino-acid frequencies, dipeptide frequencies, sequence-composition features and predicted disorder features into a single feature set, to test whether complementary representations improve on any one of them alone.

Annotated feature set

A pre-existing annotated feature set supplied within the group ("Simon's feature set", 900+ features) was also used. Missing (NaN) values in this set were identified and replaced by mean imputation so that all annotated features could be carried into the initial models.

ESM embeddings

For the PADI-score work, manually engineered features were replaced by embeddings from the ESM protein language model — a representation learned from large-scale protein sequence data rather than selected by hand.

4.3 Modeling and evaluation

Models were trained on a training split and evaluated on held-out test data. Classification results are reported as accuracy, per-class precision, recall and F1-score with support, and confusion matrices. Regression results are reported as R^2, MAE, RMSE and Spearman rank correlation, and — for the PADI work — as the percentage of predictions falling within fixed error-tolerance bands of the experimental value.

4.4 Interpretation methods

Two attribution methods were used throughout. Random Forest impurity-based feature importance gives a fast ranking of features by their contribution to the trees' splits. SHAP (SHapley Additive exPlanations) was used for the more detailed view: it assigns each feature a signed contribution to each individual prediction, which makes it possible to read not only which features matter most but in which direction a high or low value of that feature pushes the prediction. In the beeswarm plots reproduced in this report, each dot is one sequence, colour encodes whether that sequence's value for the feature is high (red) or low (blue), and horizontal position is the signed SHAP value — dots to the right push the prediction toward the positive class, dots to the left toward the negative class. Two further importance strategies, univariate R^2 screening and leave-one-feature-out ablation, are described where they were used in Section 7.

5. Plant-Specific vs. Shared Activator Classification

Recorded *Mutagenesis analysis, 11 March 2026 — the first analysis in this series.*

5.1 Objective

The goal of this initial analysis was to determine whether plant-specific activators can be distinguished from activators shared between the plant and yeast datasets on the basis of sequence-derived physicochemical properties. The classification task was defined as:

- Category 0 — plant-specific activators
- Category 1 — activators shared between plant and yeast

5.2 Model and result

A Random Forest classifier was trained on the sliding-window physicochemical representation described in Section 4.2, taking the sequence-derived window features as input and the activator category as output.

5.3 Feature contribution analysis

Feature-level analysis was performed to establish which physicochemical properties contributed most strongly to the model's predictions, and in which direction. The two figures below examine features associated with classification toward Category 0 (plant-specific) and Category 1 (shared).

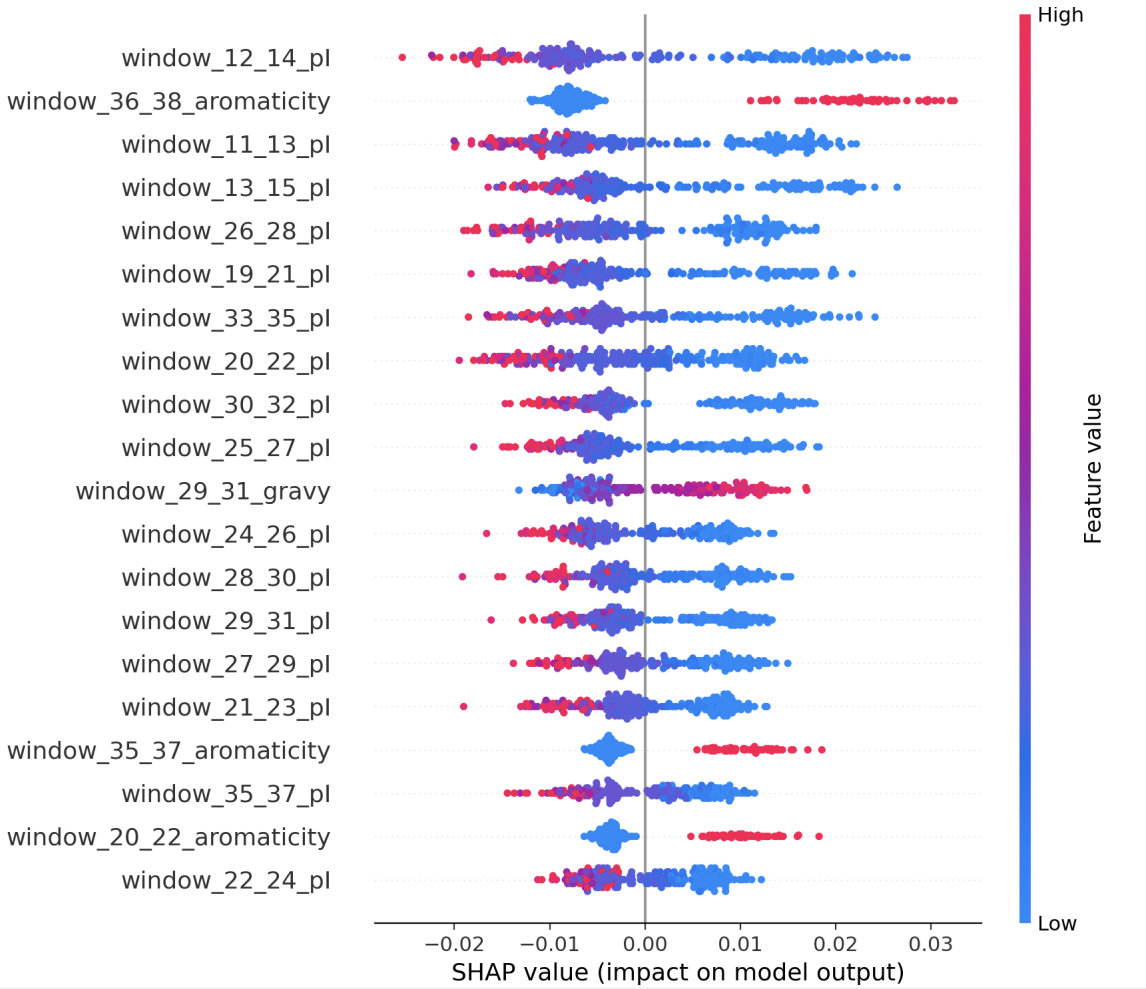


Figure 1. SHAP beeswarm for the sliding-window physicochemical Random Forest. Each row is one window-level feature (window start–end position followed by the property). Red dots are sequences with a high value of that feature, blue dots a low value; position on the x-axis is the signed contribution to the prediction, with values to the right pushing toward Category 1 (shared plant/yeast) and to the left toward Category 0 (plant-specific).

Reading the isoelectric-point windows: a high isoelectric point — that is, a high pH at which the peptide carries no net charge — pushed predictions toward Category 0, plant-specific activators. A low isoelectric point, for example in windows spanning positions 12, 13 and 14, pushed predictions toward Category 1, activators shared between plant and yeast.

Reading the other properties: high aromaticity in windows spanning positions 20–22 and 35–37 was associated with classification as a shared plant-and-yeast activator, as was high GRAVY in the window spanning positions 29–31.

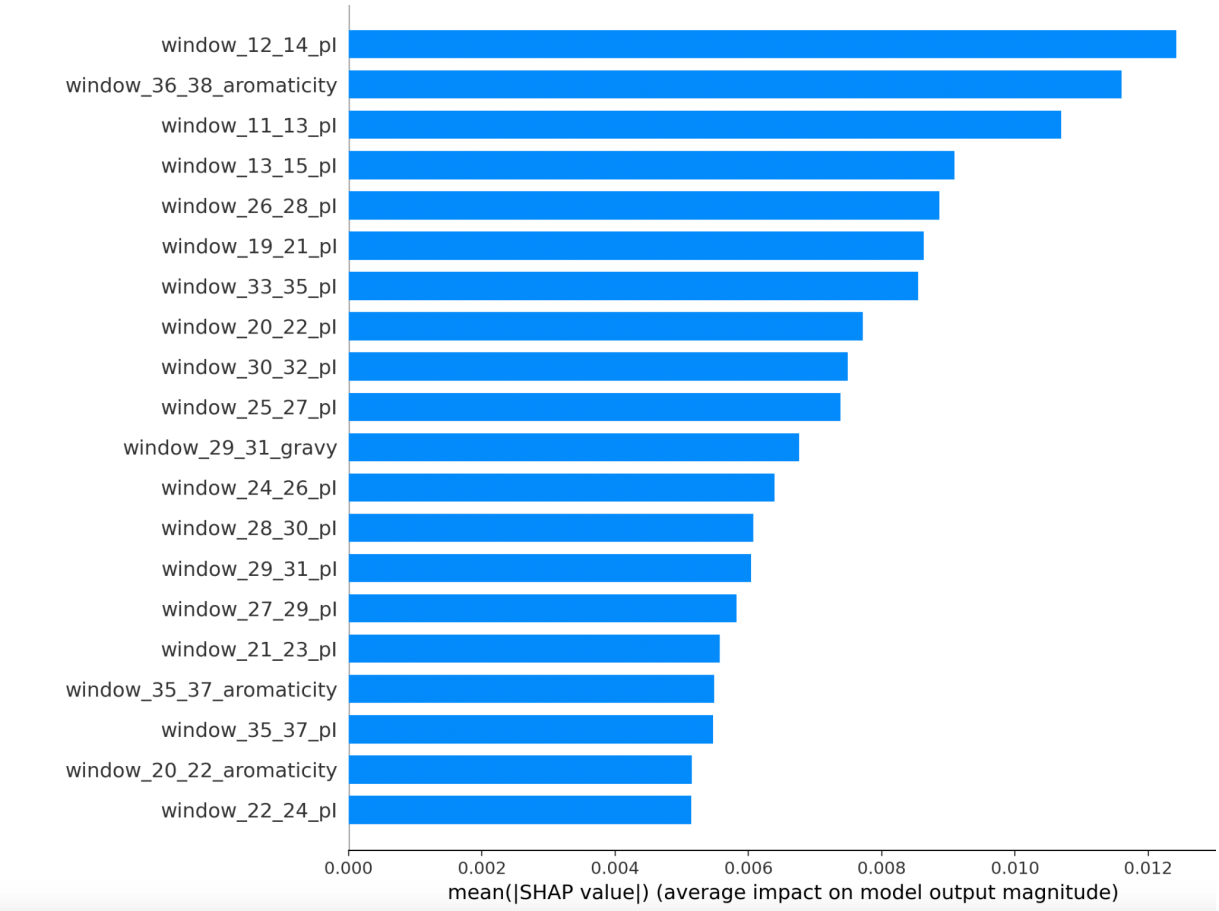


Figure 2. Mean absolute SHAP value per feature for the same model — the features with the largest overall impact on the Category 0 / Category 1 decision, irrespective of direction. Isoelectric point in the window spanning positions 12–14 ranks highest, followed by aromaticity in the window spanning 36–38.

Taken together, these two views give an initial indication of which local physicochemical characteristics differentiate plant-specific regulatory activity from activity that is conserved or shared across the plant and yeast datasets: local charge character (isoelectric point) and local aromatic content dominate the ranking, with hydropathy contributing in one specific window.

6. Comparing Feature Representations

Having established that the two activator groups are partially separable, the next question was which representation of the same protein sequences carries that signal most strongly. Four distinct feature-engineering pipelines were developed and implemented, and a separate Random Forest classifier was trained on each. Because the underlying sequences and labels are identical across the four, the comparison isolates the contribution of the representation itself.

6.1 The four models

Model 1 — all features. Every available feature combined.

Model 2 — physicochemical representation. A feature-extraction pipeline converting sequences into numerical representations of their physicochemical properties, computed with BioPython: hydrophobicity, isoelectric point, GRAVY, aromaticity, instability index and secondary-structure-related properties.

Model 3 — amino-acid composition and dipeptides. A representation based on composition and short-range patterning: individual amino-acid frequencies, dipeptide frequencies, and frequencies of selected amino acids and sequence patterns, testing whether local sequence composition alone differentiates the two categories.

Model 4 — integrated representation. Physicochemical properties, amino-acid frequencies, dipeptide frequencies, sequence-composition features and predicted disorder features combined, to test whether complementary characteristics improve classification from annotated_features.

6.2 Model 1 — all features combined

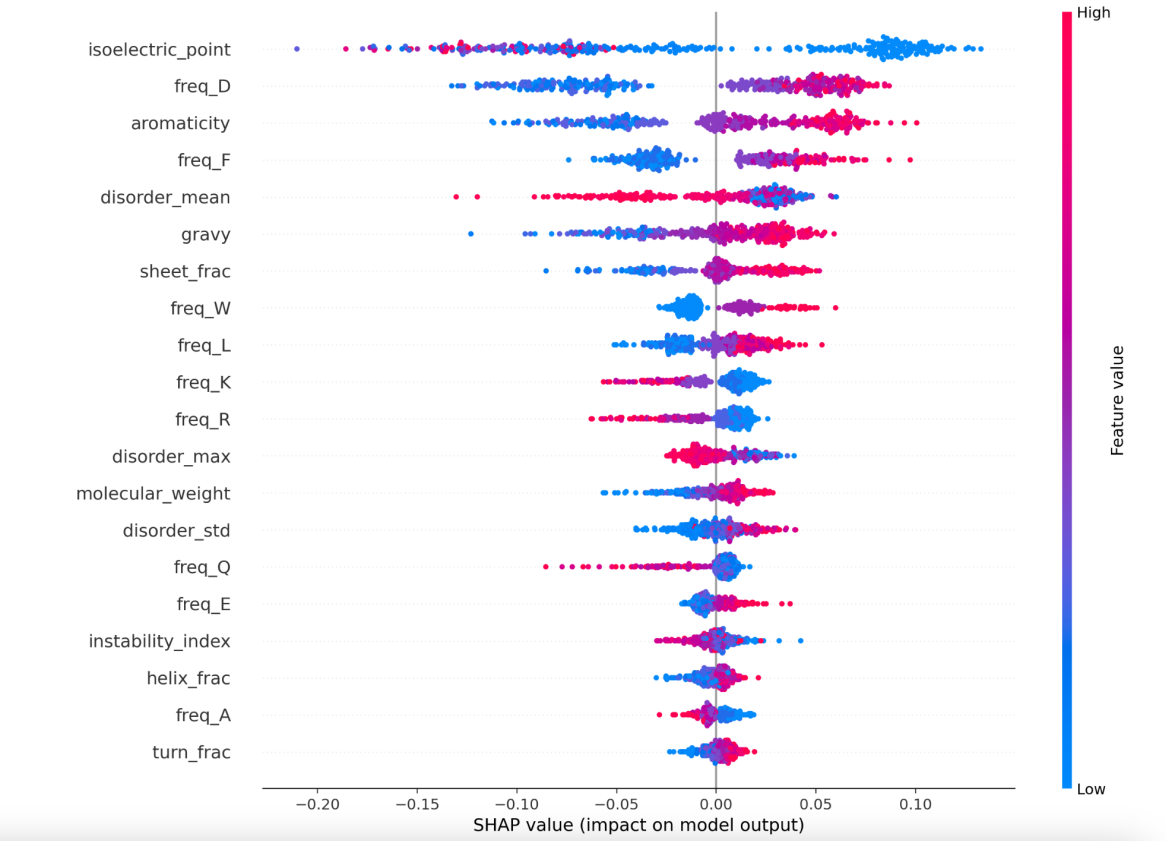


Figure 3. Model 1 (all features): SHAP beeswarm. Isoelectric point dominates, followed by the frequency of aspartate (freq_D), aromaticity, the frequency of phenylalanine (freq_F) and mean predicted disorder. Note that high mean disorder (red) sits to the left of zero, i.e. contributes toward Category 0.

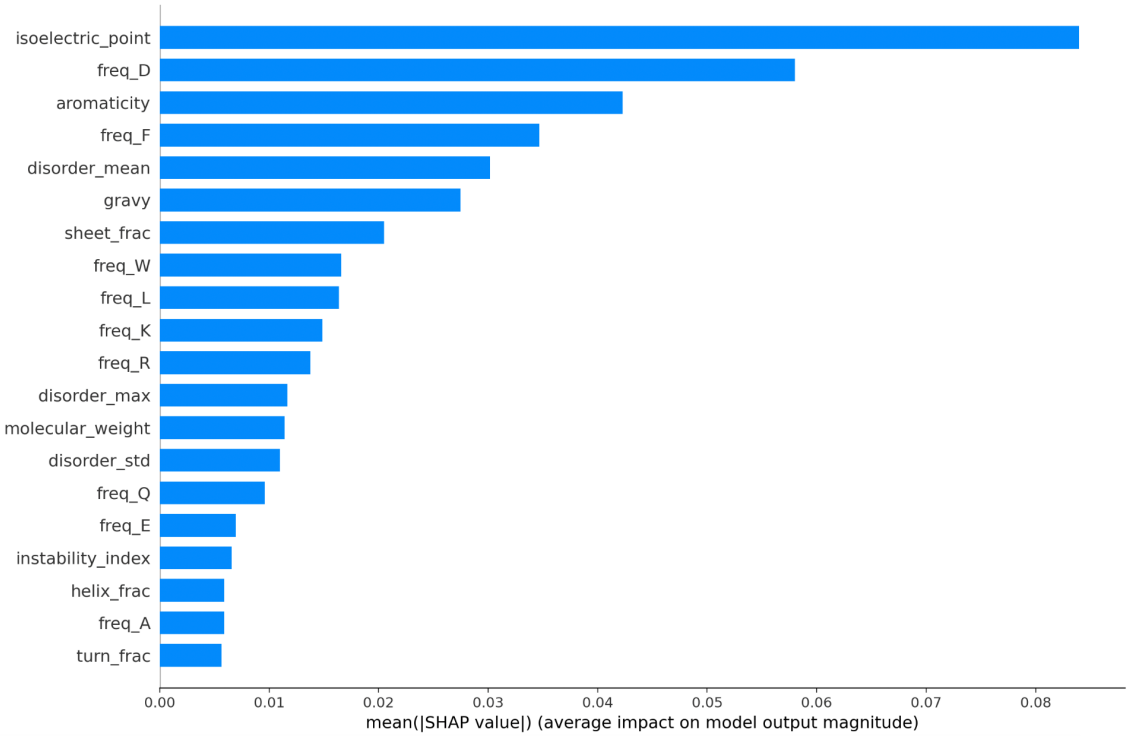


Figure 4. Model 1 (all features): mean absolute SHAP ranking of the same features, ordered by overall impact on the model output.

The combined model draws on all three families of representation at once: a physicochemical property (isoelectric point) ranks first, single amino-acid frequencies (D, F, W, L, K, R) appear throughout, and the engineered disorder summaries (mean, maximum, standard deviation) all appear in the top twenty. This is the clearest single piece of evidence that the representations are complementary rather than redundant.

6.3 Dipeptide enrichment analysis

Alongside the classifiers, a direct enrichment analysis was programmed to find the amino-acid pairs that characterise each group without going through a model at all. The procedure was:

1. Break each sequence into adjacent amino-acid pairs, separately for each of the two groups.
2. Count the number of times each pair occurs.
3. Count the total number of pairs.
4. Compute each pair's frequency as count divided by total.
5. Export the resulting values to file.
6. Plot the most shared-enriched pairs and the most plant-only-enriched pairs.

A heatmap was built to display the full 20×20 matrix of pair enrichments.

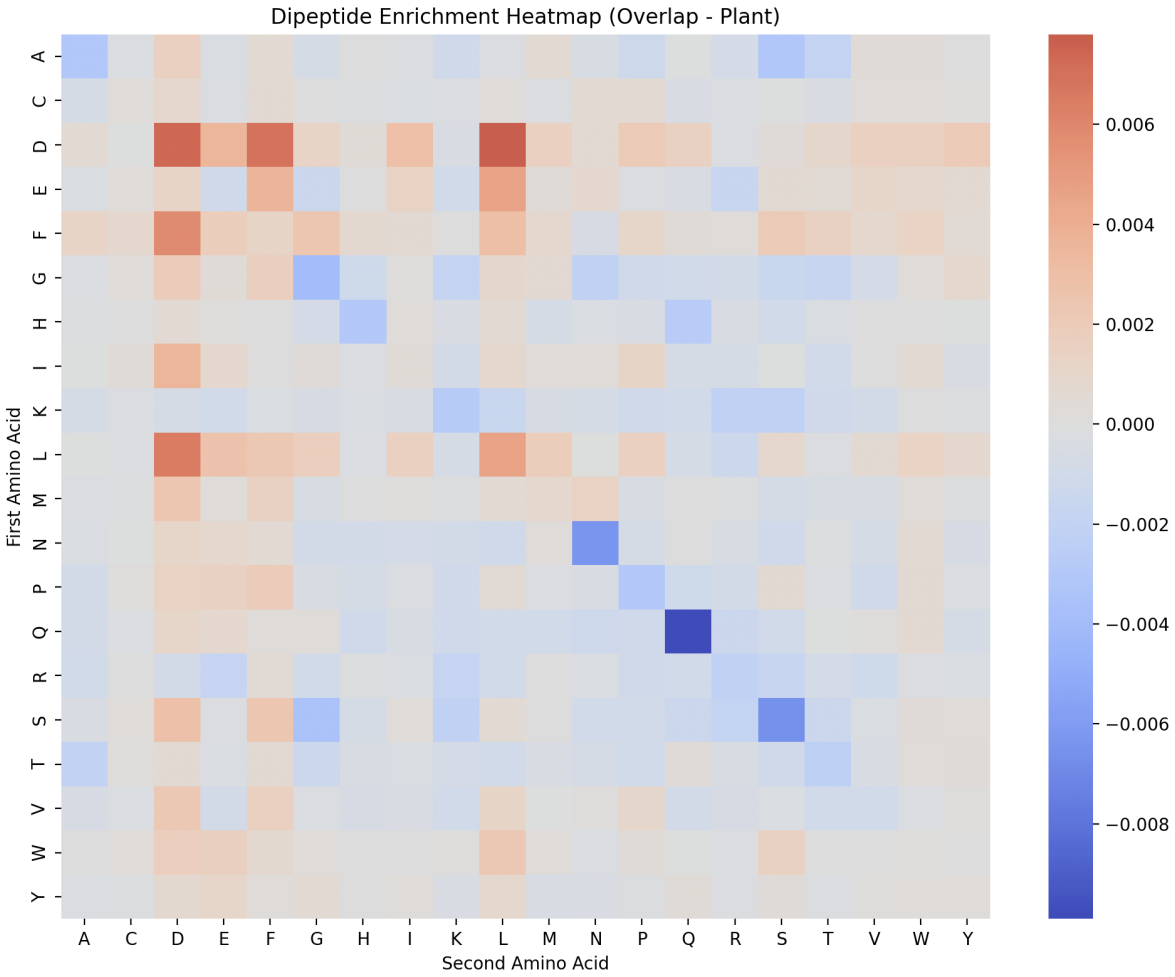


Figure 5. Dipeptide enrichment heatmap, computed as the frequency in the shared (overlap) set minus the frequency in the plant-only set. Rows are the first amino acid of the pair, columns the second. Red indicates pairs enriched in the shared activators; blue indicates pairs enriched in the plant-specific activators.

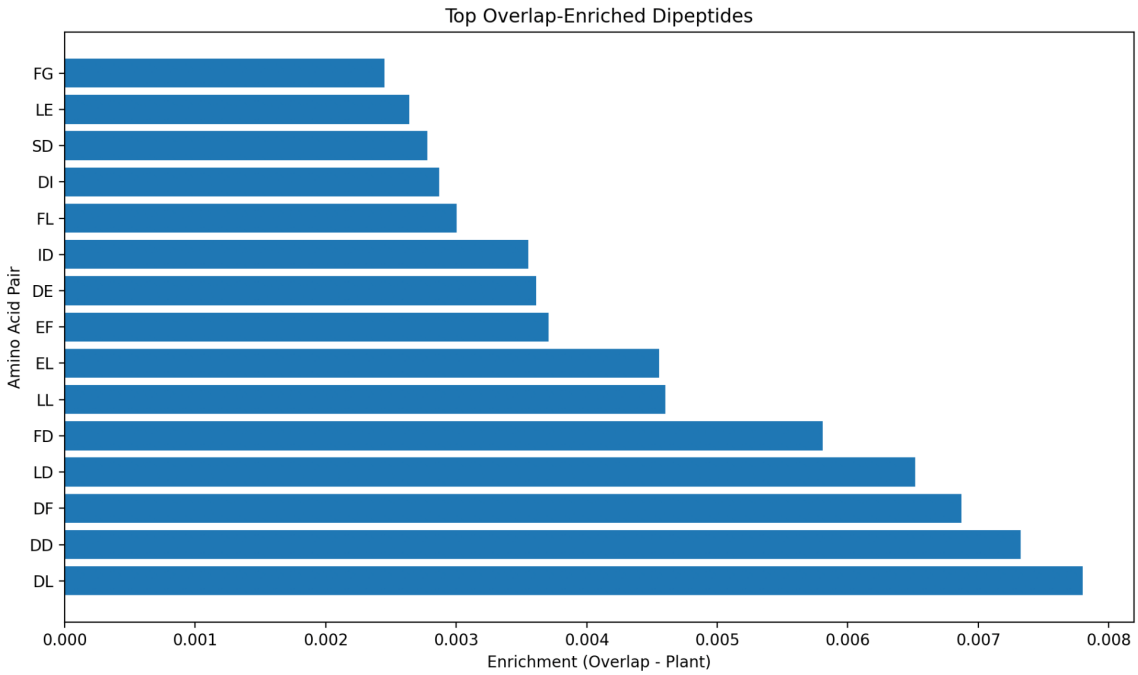


Figure 6. The most shared-enriched dipeptides, ranked by enrichment (overlap minus plant). The ranking is led by DL, DD, DF, LD and FD — pairs combining aspartate with leucine, phenylalanine or itself.

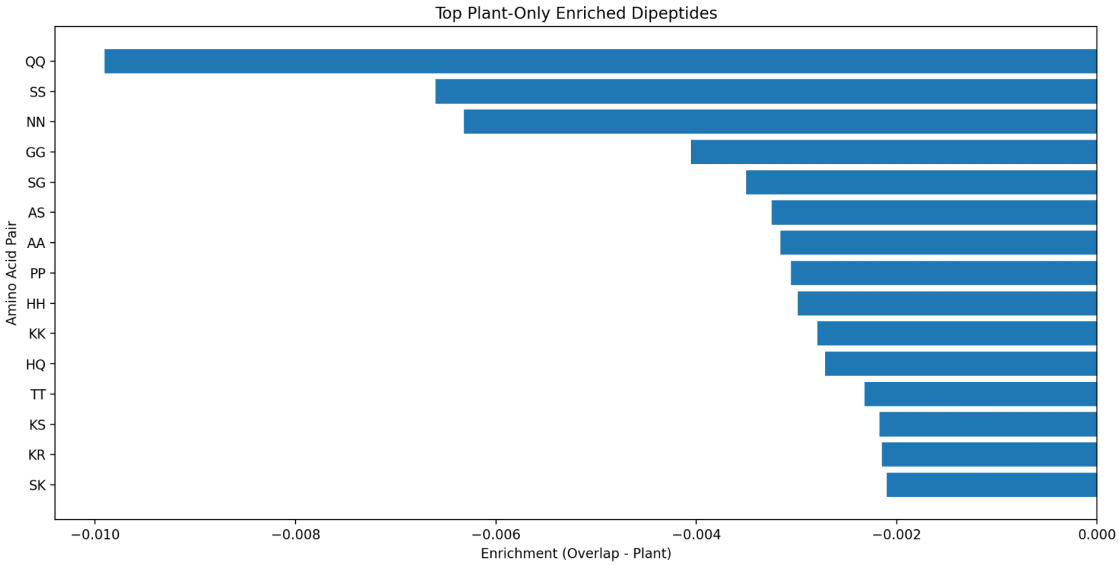


Figure 7. The most plant-only-enriched dipeptides, on the same enrichment scale (negative values indicate enrichment in the plant-specific set). The ranking is led by QQ, SS, NN, GG and SG — homotypic pairs of polar and small residues.

The two rankings point in visibly different chemical directions: shared activators are enriched in acidic–hydrophobic and acidic–aromatic pairs, while plant-specific activators are enriched in repeats of polar and small residues. This is consistent with the classifier attributions in Section 6.2, where aspartate and phenylalanine frequencies and aromaticity carried substantial weight.

6.4 Unsupervised clustering and PCA

To check whether the two groups separate without being told the labels, an unsupervised analysis was run in parallel with the supervised models. Principal component analysis (PCA) is a dimensionality-reduction technique: it re-expresses many correlated features as a small number of uncorrelated components ordered by how much of the total variation each explains, so that a high-dimensional feature set can be plotted in two dimensions. k-means clustering groups the data points by similarity, again without reference to the labels. The steps were:

- Convert each sequence to a vector of physicochemical and other training properties.
- Scale the features.
- Apply k-means clustering to group the sequences by similarity.
- Compare the resulting clusters against the actual labels.

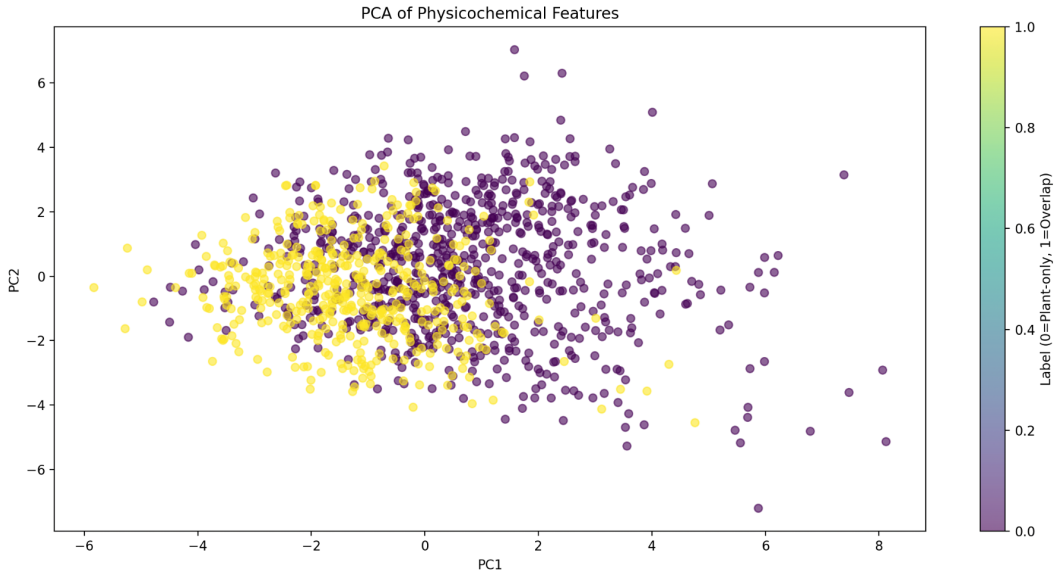


Figure 8. PCA of the physicochemical feature set, with points coloured by their true label (dark = plant-only, yellow = shared/overlap). The two classes overlap substantially in the first two principal components, while still showing some regional structure — yellow concentrating toward negative PC1, dark extending further along positive PC1.

The classes overlap but still form partly distinct regions, which suggests genuine variance in how well these features separate the two groups — consistent with a supervised accuracy meaningfully above chance but well short of clean separation.

6.5 Physicochemical properties only

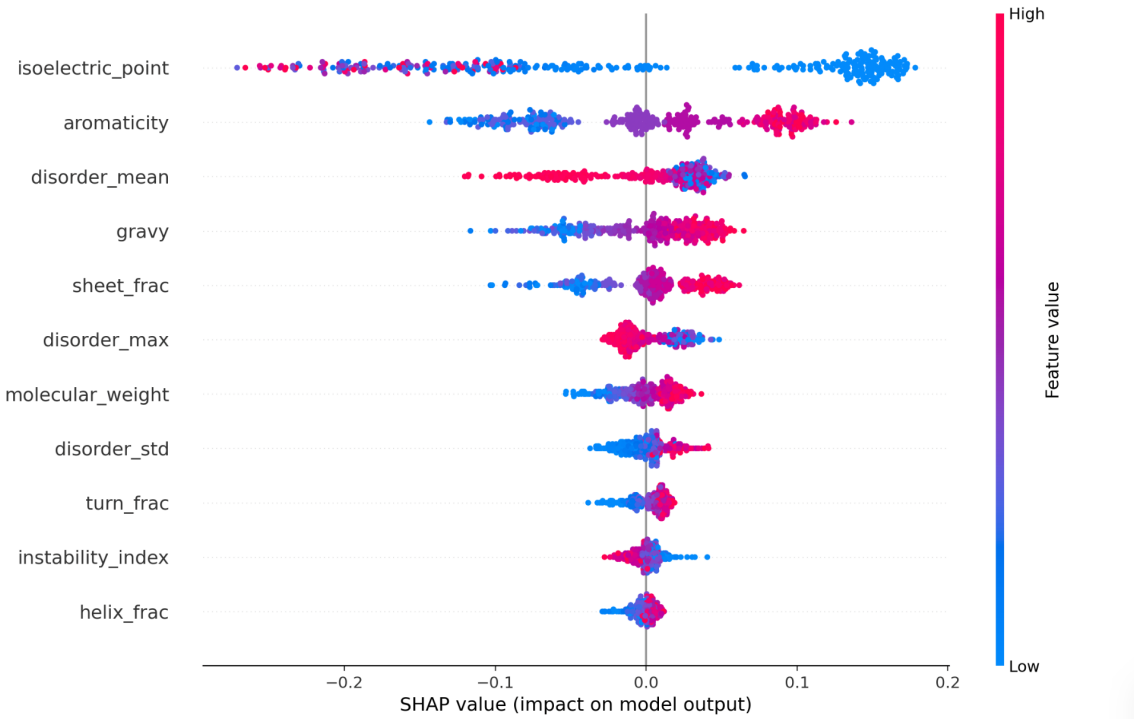


Figure 9. Model 2 (physicochemical properties only): SHAP beeswarm. Isoelectric point again dominates, followed by aromaticity, mean predicted disorder, GRAVY and sheet fraction.

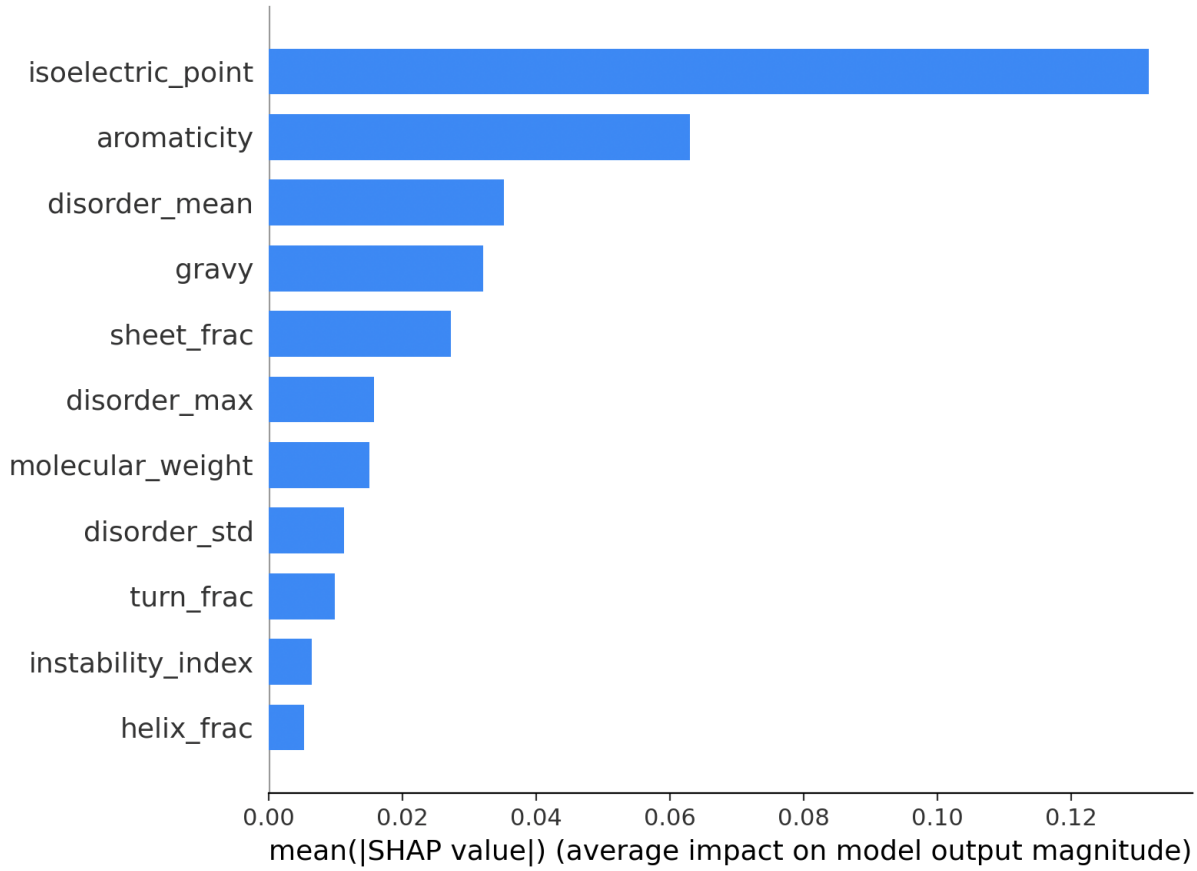


Figure 10. Model 2: mean absolute SHAP ranking. Isoelectric point contributes roughly twice as much as the next feature, aromaticity.

6.6 Dipeptides only

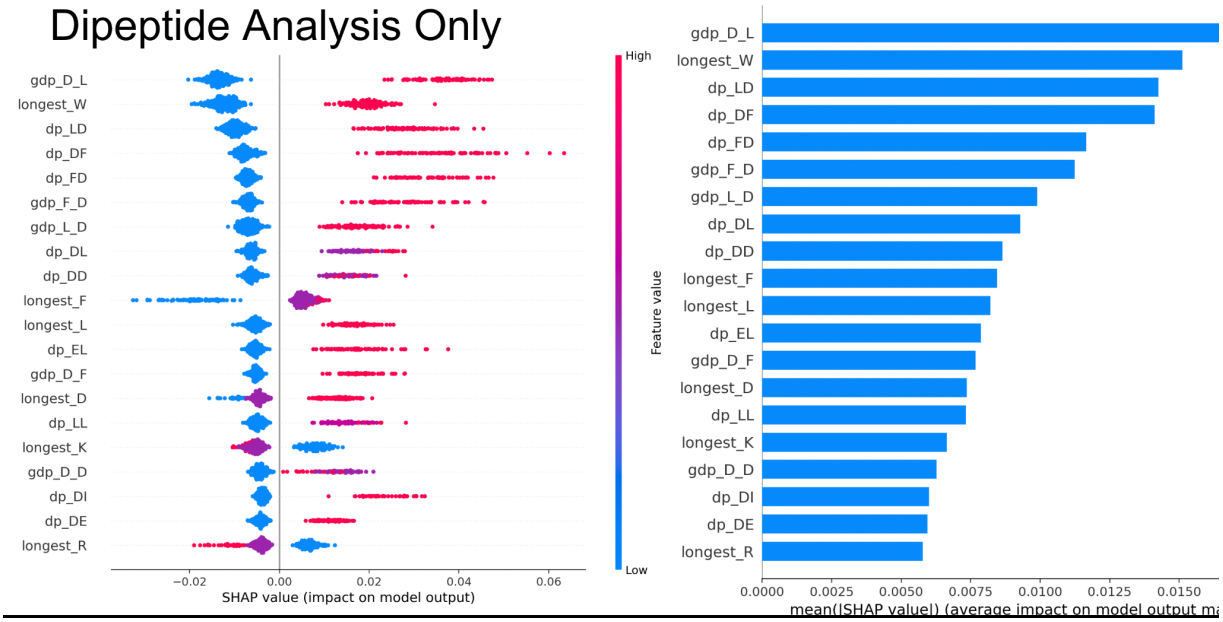


Figure 11. Model 3, dipeptide component only: SHAP beeswarm (left) and mean absolute SHAP ranking (right). Features are gapped dipeptides (gdp_), direct dipeptides (dp_) and longest homopolymeric runs (longest_). The gapped D–L pair, the longest tryptophan run and the LD, DF and FD dipeptides lead the ranking.

6.7 Amino-acid frequencies only

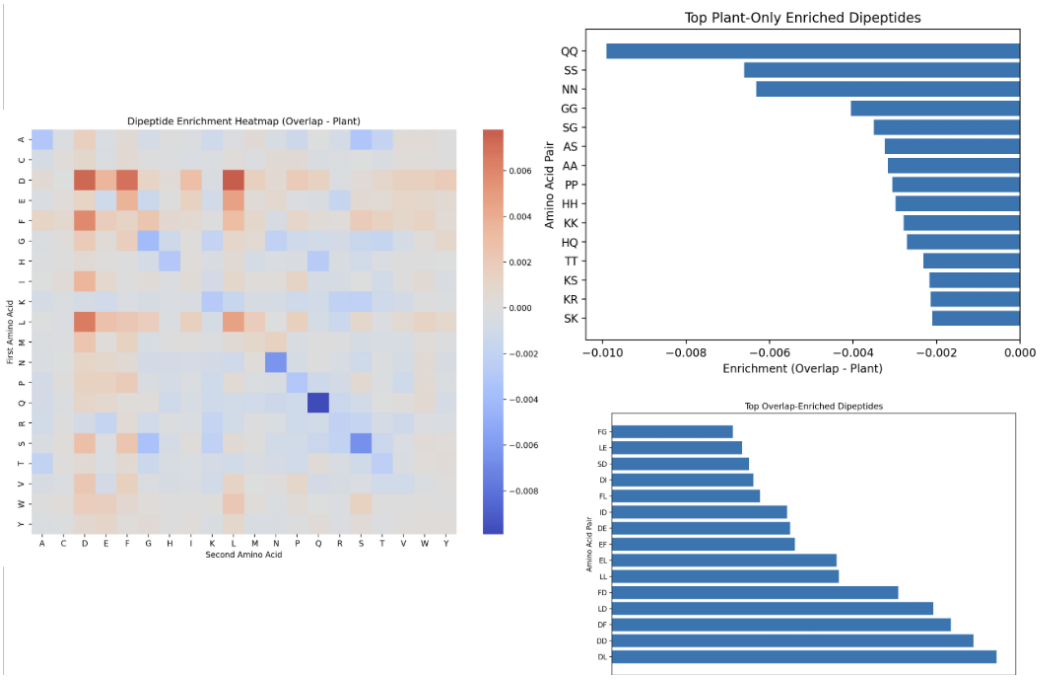


Figure 12.

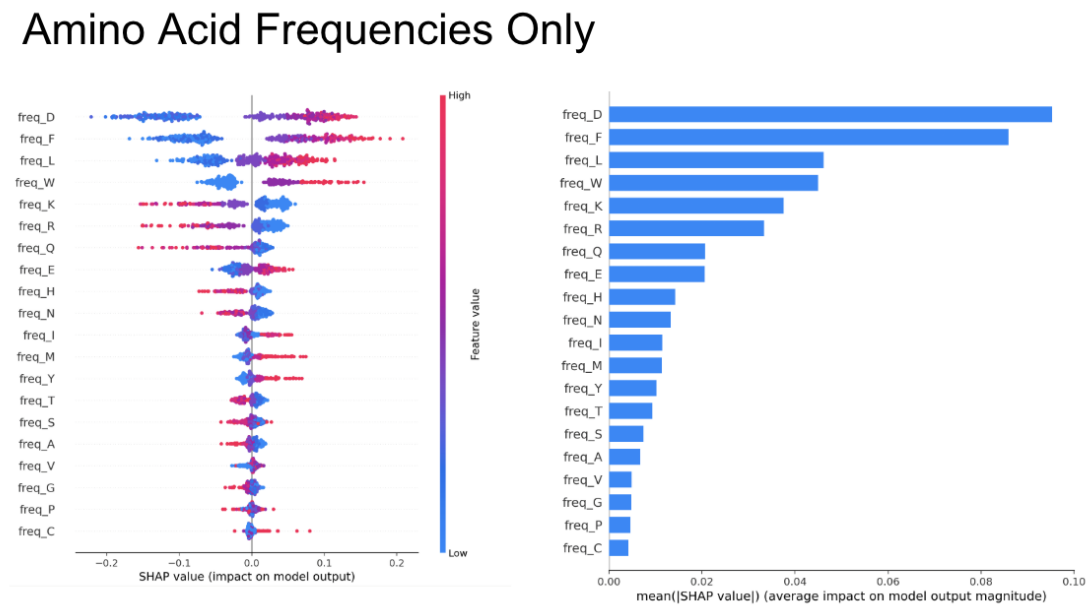


Figure 13. Model 3, amino-acid frequency component only: SHAP beeswarm (left) and mean absolute SHAP ranking (right). The frequencies of aspartate (D) and phenylalanine (F) dominate, followed by leucine (L) and tryptophan (W); the basic residues lysine (K) and arginine (R) contribute in the opposite direction.

Because the amino-acid frequency features carried so much of the signal, the distributions of the three highest-ranked frequencies were plotted directly for the two groups, to confirm that the model was responding to a real distributional difference rather than to an artifact of the fit. A visible difference is present in all three.

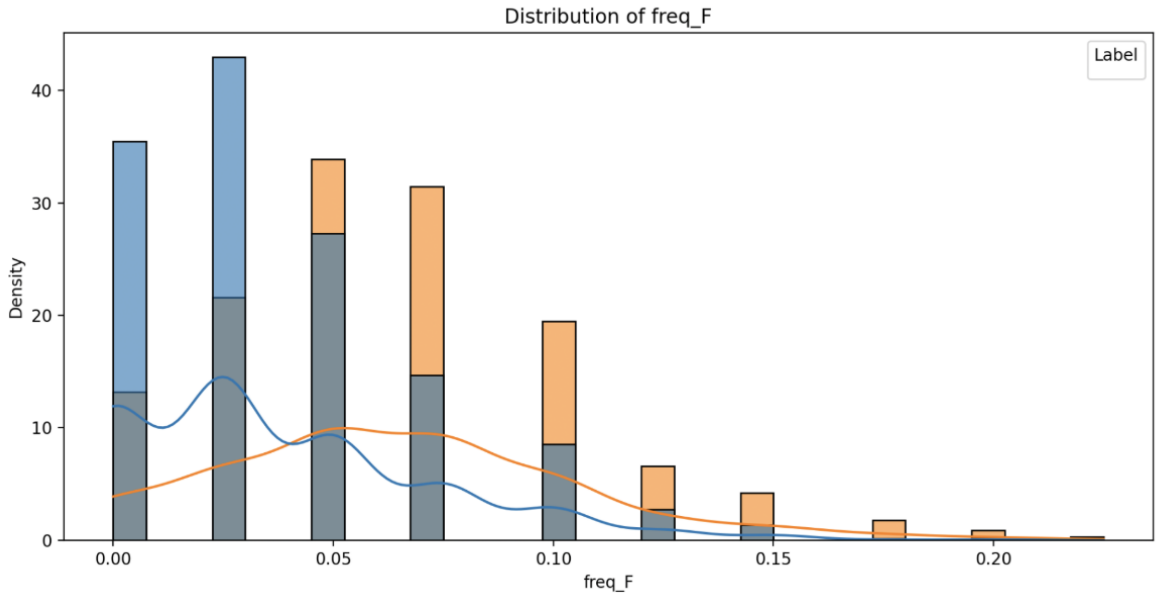


Figure 14. Distribution of phenylalanine frequency (freq_F) by group. High values are predominantly orange — the shared plant-and-yeast activators — while the plant-specific set (blue) concentrates at low values.

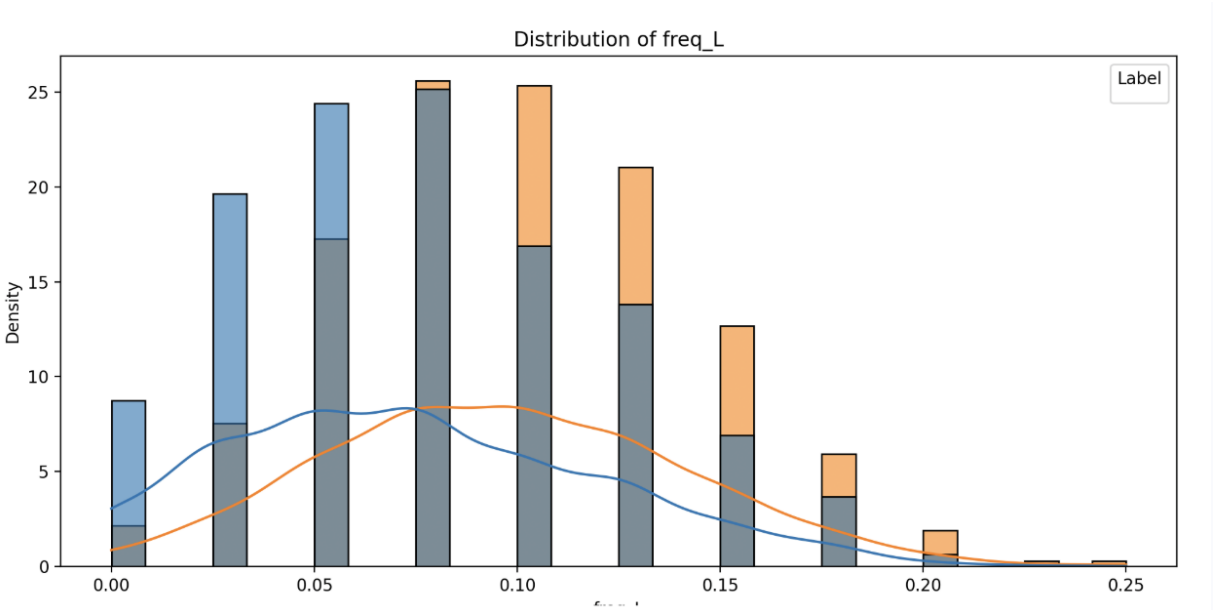


Figure 15. Distribution of leucine frequency (freq_L) by group, showing the same pattern: the orange (shared) distribution is shifted toward higher values than the blue (plant-specific) one.

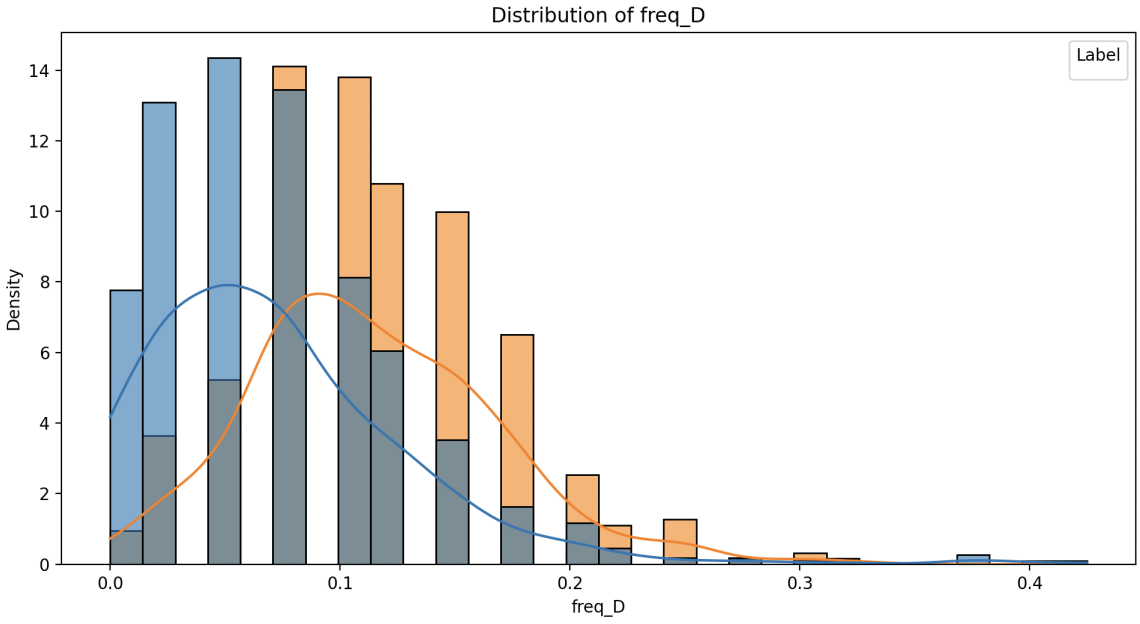


Figure 16. Distribution of aspartate frequency (freq_D) by group, again shifted toward higher values in the shared set.

6.8 Random Forest on the annotated feature set

A separate Random Forest pipeline was developed and trained on the annotated feature set given by Simon. Several modeling and feature-selection approaches were tested before settling on the following final architecture.

1. Missing-value imputation — NaN feature values were identified and replaced by mean imputation, allowing all annotated features into the initial model.
2. Initial Random Forest — trained on the complete annotated feature set.
3. Feature selection — the initial model's feature-importance scores were used to rank the annotated features by their contribution, and the feature space was reduced to the top 100.
4. Final Random Forest — a second classifier trained on the selected top 100 features only, reducing the feature space while retaining what the first model identified as most informative.
5. Evaluation — the final classifier was evaluated on held-out test data.
6. SHAP analysis — performed on the final model to give an interpretable view of which annotated features contributed most strongly, the relative magnitude of their contributions, and which features pushed predictions toward each category.

This pipeline was applied to two further activator-group comparisons, with the results below.

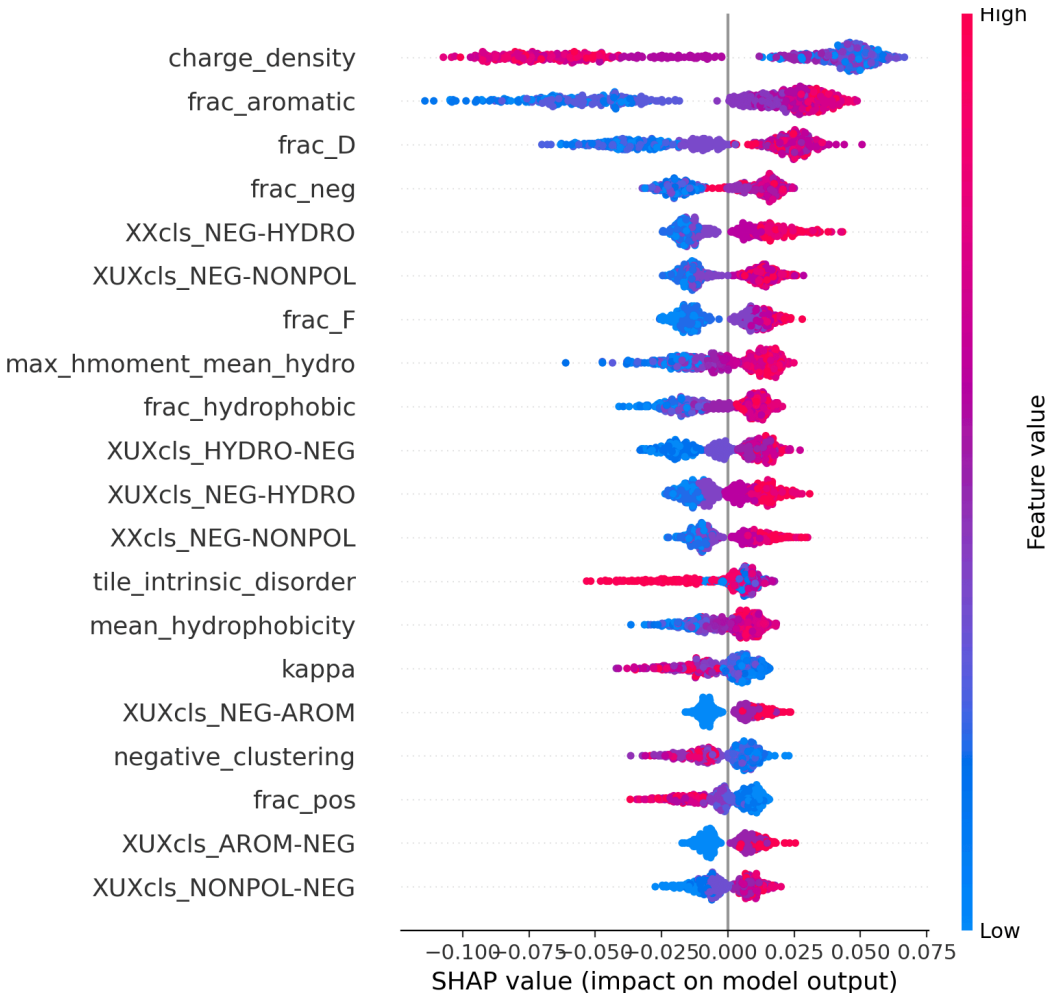
Table 1 Model 4:

Comparison	Feature set	Model	Accuracy
Plant-specific vs. shared	Annotated feature set	Random Forest	81.935%
Yeast-specific vs. plant-specific	Annotated feature set	Random Forest	72.00%
Yeast-specific vs. shared	Annotated feature set	Random Forest	79.89%

Plant-specific vs shared

Accuracy: 0.81935% (81.935%)

```
Confusion Matrix:
[[301  52]
 [ 60 207]]
```



Class	Precision	Recall	F1-score
0	0.83	0.85	0.84
1	0.80	0.78	0.79

Yeast-specific vs. plant-specific activators

Accuracy: 0.7200 (72.00%)

Confusion matrix

```
[[116 204]
 [ 74 599]]
```

Table 2. Yeast-specific vs. plant-specific activators — classification report. Class 0 is plant-specific, class 1 yeast-specific. Recall on class 0 is low (0.37): the model identifies yeast-specific activators well but misses roughly two-thirds of the plant-specific ones.

Class	Precision	Recall	F1-score	Support
0	0.61	0.36	0.45	320
1	0.75	0.89	0.81	673

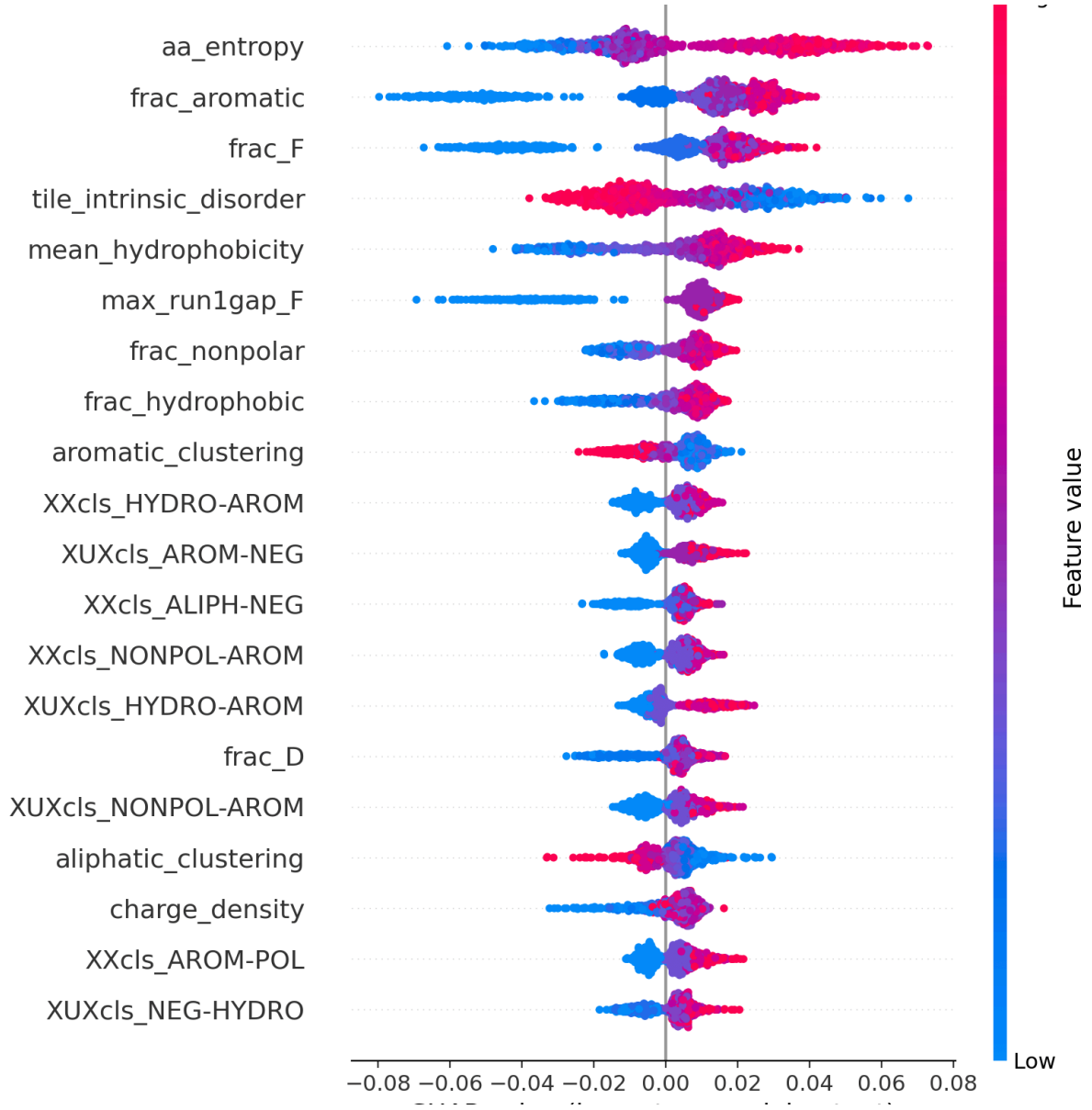


Figure 17. SHAP beeswarm for the yeast-specific vs. plant-specific classifier on the annotated feature set. Entropy, Aromatic fraction lead the ranking.

Yeast-specific vs. shared activators

Accuracy: 0.8138 (81.38%)

Confusion matrix

```
[[590  83]
 [106 161]]
```

Table 3. Yeast-specific vs. shared activators — classification report. Class 0 is yeast-specific, class 1 shared. This is the best-separated of the three activator-group comparisons, though the smaller shared class is still the harder one to recall.

Class	Precision	Recall	F1-score	Support
0	0.85	0.88	0.86	673
1	0.66	0.60	0.63	267

Across the three comparisons a consistent pattern emerges: the majority class in each split is recalled well and the minority class poorly. This is the same class-imbalance behavior that became the central obstacle in the next workstream.

7. Plant Activator vs. Non-Activator Modeling

7.1 Objective and initial model

A machine-learning pipeline was developed to distinguish plant activators from plant non-activators using annotated sequence features. Preprocessing scripts were written to construct the classification dataset, with class 1 as plant activators and class 0 as plant non-activators, and an initial Random Forest classifier was built and trained from scratch.

Initial evaluation revealed a substantial difference in performance between the two classes with a <20% recall for the activator class and an extremely high (>99%) recall on non activator classes.

The particularly low activator recall showed that the model was strongly biased toward predicting the majority non-activator class. Predicted activators were often correct, but the model failed to identify a large fraction of the true activators. Subsequent experiments therefore focused on improving minority-class recall while maintaining acceptable precision.

7.2 Class-imbalance mitigation

Class weighting

The relative class weights were modified to penalize incorrect predictions of the minority activator class more heavily, at 2:1 and at 3:1 toward the minority class. Increasing the minority-class weight shifted the classifier's behavior, but precision and recall traded off against one another rather than producing a substantial overall improvement.

Activator subset experiments

Training was also restricted to the top 75% of activators, to test whether removing potentially weaker or noisier positive examples would improve class separation. This produced little change in activator precision or recall.

Balanced Random Forest and hyperparameter optimization

Random Forest, Balanced Random Forest and GridSearchCV-driven systematic hyperparameter optimization were implemented and compared, and these experiments were repeated using the top-75% activator subset.

Several classifiers were additionally tested using Lucas's notebook and balanced accuracy remained at approximately 50%, comparable to the performance ceiling observed with the existing approaches.

HistGradientBoosting

Because gradient-boosting methods can capture complex nonlinear relationships between features, a HistGradientBoosting classifier was built and fine-tuned. Its overall performance remained similar to the Random Forest-based approaches, indicating that changing classifier architecture alone was not sufficient to resolve the underlying problem.

7.3 Investigating the feature space

The dataset contained more than 900 annotated features, raising the possibility that irrelevant, redundant or confounding features were limiting generalizations. The focus therefore shifted from optimizing the classifier to investigating the feature representation itself. A SHAP-based feature-selection pipeline was developed:

```
Train on all features → SHAP analysis → rank features → select top features → retrain → re-evaluate precision and recall
```

Models were separately evaluated using the top 100, top 200 and top 300 SHAP-ranked features. Reducing the feature space produced a small improvement, but not enough to resolve the low activator recall — suggesting that feature dimensionality alone was unlikely to explain the classification difficulty.

7.4 Reframing the problem as regression

Following discussions with Hector's group, the question was reframed. Rather than predicting a binary activator / non-activator label directly, the model would predict the underlying continuous enrichment value, and the classification boundary would be applied afterwards. This changed the problem from:

```
Sequence features → activator / non-activator
```

to:

```
Sequence features → predicted enrichment value → activator threshold
```

The advantage is that the model learns the continuous biological signal first and the discrete decision is imposed afterwards, where it can be varied.

Univariate feature screening and Ridge regression

Following Professor Max Staller's suggestion to Simon to evaluate features individually, a three-stage feature-screening and regression pipeline was designed and implemented.

Stage 1 — individual feature evaluation. For each annotated feature: train a univariate Linear or Ridge regression using that feature alone; predict the continuous enrichment value; calculate the resulting R^2; and rank features by their individual explanatory power. This gives a systematic measure of which single features are most strongly associated with enrichment.

Stage 2 — multivariate Ridge regression. The top 100 features from the univariate screen were used to train a multivariate Ridge regression predicting enrichment.

Stage 3 — threshold-based classification. Rather than fixing a single boundary, several thresholds on the predicted enrichment value were evaluated — 0.5, 1.0, 1.5, 2.0 and 2.5 — with predicted enrichment at or above the threshold classed as activator and below it as non-activator. This shows how different biological decision boundaries affect the resulting precision and recall.

Metric	Value
Train R^2	0.1403
Test R^2	0.1379
MAE	0.4673

Metric	Value
RMSE	0.6716
Spearman correlation	0.4015

The top-ranked features from the univariate screen were recorded as follows.

#	Feature
0	charge_density
1	frac_pos
2	frac_R
3	XXcls_POS-HYDRO
4	XXcls_HYDRO-POS
5	XXcls_POS-NONPOL

7.5 Leave-one-feature-out analysis

An alternative feature-importance strategy was designed and implemented, based on ablation rather than attribution. For each feature: remove it from the dataset; retrain and evaluate the model; measure the resulting change in accuracy, precision and recall; and use that change to estimate the removed feature's importance. Features were then ranked by their effect on performance, the top 100 selected, that set used to predict enrichment, and manual enrichment thresholds applied to generate activator / non-activator predictions. This gives a measure of importance grounded in the actual change in predictive performance when a feature is withheld, which is a stricter test than impurity- or SHAP-based ranking on a single fitted model.

7.6 Additional regression architectures

Following a second discussion with Juan and Jacob, the regression pipeline was expanded to investigate further model architectures and feature-selection methods.

Lasso regression

Lasso regression was incorporated to introduce L1 regularization and encourage sparse feature selection in the high-dimensional feature space. SHAP analysis was then performed directly on the regression model to identify which features contributed most strongly to predicted enrichment. The resulting pipeline was:

```
Annotated features → Lasso regression → enrichment prediction → SHAP
→ feature ranking → threshold classification
```

Extra Trees regression

An Extra Trees Regressor was also built and trained, to capture potentially nonlinear relationships and interactions between annotated features and enrichment. This provided a nonlinear alternative to Ridge and Lasso and allowed a direct comparison between linear regularized models and tree-based ensemble regression.

Handover ***These were the best results obtained on this task. Because the right way to proceed was not clear, this model was handed to Juan and Jacob rather than developed further.***

8. Further Comparisons

8.1 Plant + yeast activators vs. plant non-activators

A series of Random Forest classification models was developed and trained to investigate sequence-level differences between activators identified across the plant and yeast datasets and plant non-activators. Three separate models were built, each trained on a different feature representation of the same protein sequences, and each interpreted through SHAP.

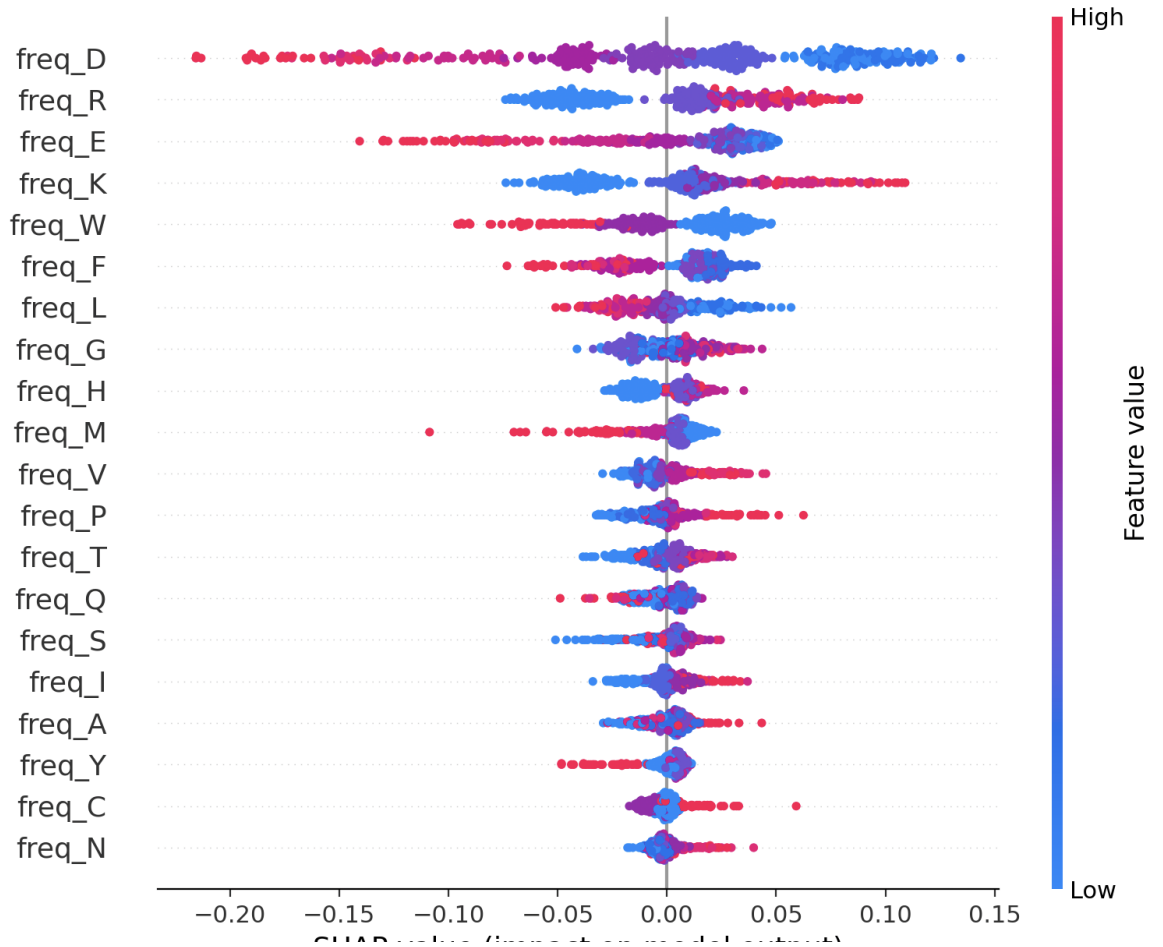


Figure 18. Amino-acid frequency representation only: SHAP beeswarm. The frequency of aspartate (freq_D) dominates, with high values (red) contributing toward the negative side and low values (blue) toward the positive; the frequencies of arginine, glutamate, lysine and tryptophan follow.

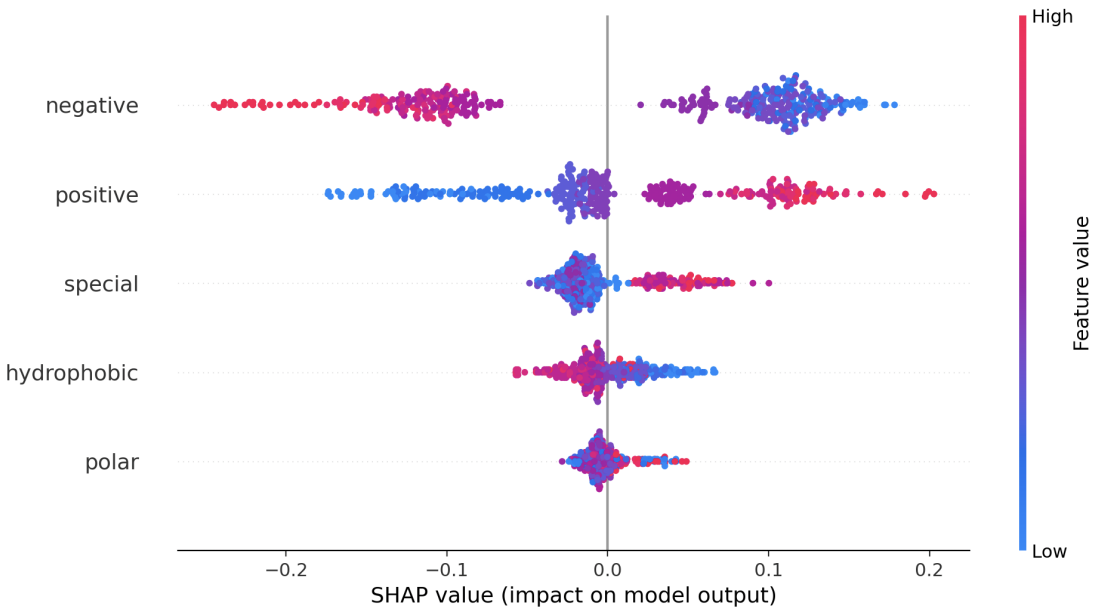


Figure 19. Amino-acid composition-class representation only: SHAP beeswarm over five residue classes. The negative and positive classes carry almost all of the signal and act in opposite directions — a high fraction of negatively charged residues pushes one way, a high fraction of positively charged residues the other — while the special, hydrophobic and polar classes contribute comparatively little.

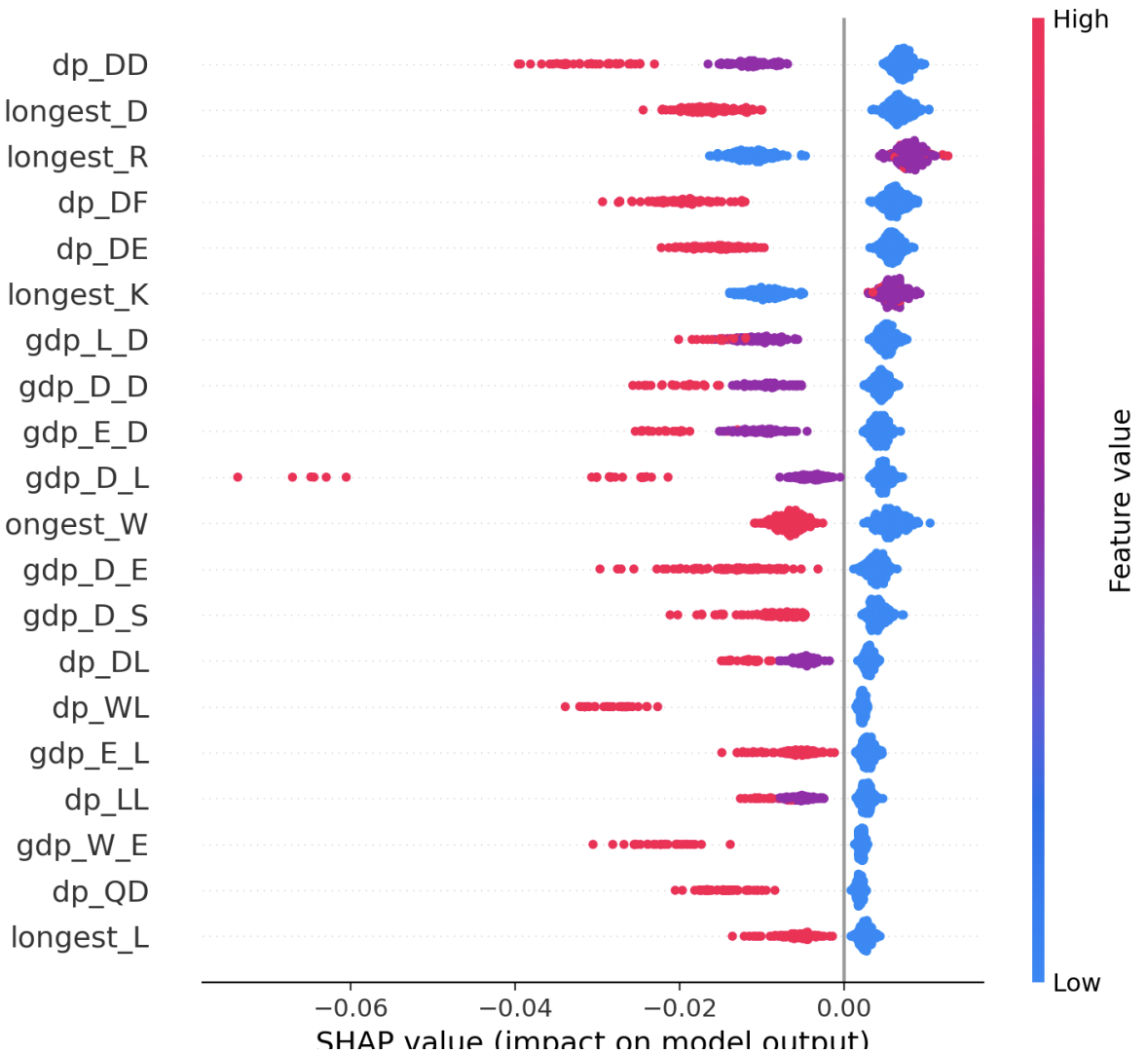


Figure 20. Dipeptide representation only: SHAP beeswarm over direct dipeptides (dp_), gapped dipeptides (gdp_) and longest homopolymeric runs (longest_). The DD pair, the longest aspartate run and the longest arginine run lead the ranking.

Read together, the three representations converge on the same chemistry from three directions: charge. The frequency model puts aspartate first, the composition-class model reduces almost the entire signal to the negative and positive residue fractions acting in opposition, and the dipeptide model leads with aspartate pairs and aspartate runs. That the same conclusion survives three independent encodings of the sequence is a stronger result than any single ranking.

8.2 Plant repressors vs. non-repressors

Rationale and batch-based strategy

Following discussion with Simon, he mentioned that because the main question here was which features contribute most rather than the predictive power of any single overall model — and because of the imbalance in the dataset — a batch-based strategy was chosen.

To prevent the larger non-repressor population from dominating the analysis, multiple class-balanced batches were constructed, each containing equal numbers of repressors and non-repressors. Across batches:

- The repressor set was held constant.
- A new set of non-repressors was randomly sampled for each batch.
- Each batch contained equal numbers of repressors and non-repressors.
- A separate Random Forest classifier was trained on each batch.

This makes it possible to examine whether the same features repeatedly emerge as important across different samples of the non-repressor population. Rather than relying on the feature importance of a single model, the stability and consistency of important features can be evaluated across independently trained models — which is the appropriate design when feature discovery, not accuracy, is the goal.

Three experimental conditions

The analysis was designed around three representations of the sequence dataset.

1. Original sequences. Random Forest models trained on features derived from the original, unmodified sequence data, establishing a baseline for identifying features associated with repressor versus non-repressor classification.

2. Motif-mutated sequences. A second condition in which known sequence motifs were randomly mutated. Models were retrained on the modified sequences to investigate whether the important features remained predictive once information from the known motifs was disrupted — that is, whether the classifier was relying primarily on known motif signals, or whether additional sequence characteristics continued to distinguish repressors from non-repressors.

3. Motif-removed sequences (in progress). A third analysis in which sequences containing specified motifs are removed from the dataset entirely, to determine whether predictive features can still be identified after excluding those sequences. This is a more stringent test of whether the models identify features associated with repressor activity beyond the predefined motifs.

9. PADI-Score Regression from Protein Sequence

As a next step toward transfer-learning-based sequence modeling, a regression pipeline was developed to predict a continuous PADI score directly from protein sequence. The overall task was:

```
Protein sequence → sequence representation → regression model → predicted PADI score
```

Engineered physicochemical features, pretrained protein-language-model embeddings, ensemble methods, neural-network architectures and linear regression approaches were compared systematically, to determine which representation and which model best capture the sequence-to-PADI relationship. Unlike the classification workstreams, this task has a continuous target throughout, so no threshold is involved and performance can be read directly as prediction error.

9.1 Physicochemical and positional feature regression

The first approach was an ensemble combining the predictions of a Random Forest Regressor and a CatBoost Regressor, trained on engineered sequence features comprising physicochemical properties and positional encoding of amino acids. The two regressors' predictions were combined to give the final predicted PADI score.

9.2 Transfer learning with ESM embeddings

The manually engineered representation was then replaced by embeddings from the ESM protein language model, which provides a representation pretrained on large-scale protein sequence data rather than one built from hand-selected physicochemical properties. ESM embeddings were extracted for the sequences and used as input to the same Random Forest and CatBoost ensemble.

Representation	Model	RMSE	MAE	R^2
Engineered physicochemical + positional	RF + CatBoost ensemble	0.772	0.429	0.389
ESM embeddings	RF + CatBoost ensemble	0.784	0.437	0.371

The limited explained variance of the first model motivated the move to richer representations; the result of that move was that the representation was not the binding constraint for this model family. That finding is what pointed the work toward changing the regressor instead, which is what Sections 9.3 to 9.5 do.

9.3 Ridge regression — best-performing model

Finally, a Ridge Regression model was developed on the ESM embeddings. Despite being considerably simpler than the nonlinear ensemble and neural-network architectures, Ridge produced the strongest results of any model tested.

Error tolerance	Ridge on ESM embeddings
±0.2	58.35%
±0.3	72.26%
±0.4	79.11%
±0.5	83.09%

10. Glossary

Definitions of the technical terms and metrics used in this report. These are standard definitions of the methods applied; they do not add findings.

Accuracy — The proportion of all predictions that are correct. Misleading under class imbalance: a model that always predicts the majority class can score highly while identifying none of the minority class.

Amino-acid frequency — The proportion of a sequence made up of each of the twenty amino acids — a fixed-length, position-independent summary of composition.

Aromaticity — The relative frequency of the aromatic residues (phenylalanine, tryptophan, tyrosine).

Balanced accuracy — The average of the recalls obtained on each class. Unlike plain accuracy it is not inflated by majority-class performance; 50% corresponds to chance on a two-class problem.

Balanced Random Forest — A Random Forest variant that under-samples the majority class when building each tree, so every tree sees a class-balanced sample.

BioPython — A Python library for biological computation, used here to calculate physicochemical sequence properties.

CatBoost — A gradient-boosting library that builds an additive ensemble of decision trees, each correcting the errors of those before it.

Class imbalance — The situation where one class greatly outnumbers another in the training data. Models trained on imbalanced data tend to favour the majority class, because doing so minimises overall error.

Class weighting — Assigning a larger penalty to errors on the minority class during training, to counteract majority-class bias.

CNN (convolutional neural network) — A neural architecture that learns local patterns via filters slid across the input — applicable to sequences as well as images.

Confusion matrix — A table of predicted against true classes. For two classes: [[true negatives, false positives], [false negatives, true positives]].

Cross-validation — Repeatedly splitting the data into training and validation folds and averaging the results, giving a more robust performance estimate than a single split.

Dipeptide frequency — The frequency of each of the 400 possible adjacent amino-acid pairs. A gapped dipeptide is a pair separated by one or more intervening residues.

Enrichment value — The continuous experimental measure of activity that underlies the binary activator label; predicting it directly, then thresholding, is the reframing described in Section 7.4.

ESM — A protein language model pretrained on large-scale sequence data. Its learned per-sequence embeddings can be used as input features in place of hand-engineered ones.

Extra Trees — Extremely Randomised Trees — a tree ensemble that chooses split thresholds at random rather than optimally, reducing variance but with a tendency to overfit in some settings.

F1-score — The harmonic mean of precision and recall, summarising both in one number.

Feature ablation / leave-one-feature-out — Estimating a feature's importance by removing it, retraining, and measuring how much performance drops.

Feature importance (impurity-based) — A Random Forest's built-in ranking of features by how much they reduce node impurity across the trees. Fast, but biased toward high-cardinality features and blind to direction of effect.

GRAVY — Grand Average of Hydropathy — the mean hydropathy value of all residues in a sequence.

GridSearchCV — Exhaustive search over a specified grid of hyperparameter values, scoring each combination by cross-validation.

HistGradientBoosting — A histogram-based gradient-boosting classifier, efficient on large feature sets and able to capture nonlinear feature interactions.

Hydrophobicity / Kyte–Doolittle scale — A per-residue scale of water-repellence; summing or averaging it over a sequence or window gives a hydropathy measure.

Instability index — A sequence-derived estimate of protein instability computed from the occurrence of particular dipeptides.

Intrinsic disorder — Regions of a protein that do not adopt a fixed three-dimensional structure. Predicted per-residue by tools such as metapredict.

Isoelectric point (pI) — The pH at which a peptide carries no net electrical charge — a summary of its charge composition.

k-means clustering — An unsupervised method that partitions data into k groups by similarity, without reference to labels. Comparing clusters against true labels tests whether class structure exists in the feature space on its own.

Lasso regression — Linear regression with an L1 penalty, which drives some coefficients exactly to zero and therefore performs feature selection as part of fitting.

MAE (mean absolute error) — The average absolute difference between predicted and true values, in the units of the target.

Mean imputation — Replacing missing values with the mean of the observed values for that feature, so that rows with gaps need not be discarded.

metapredict — A disorder-prediction tool returning one disorder score per residue. Because that output varies in length with the sequence, it must be summarised (mean, standard deviation, maximum) to serve as fixed-length features.

MLP (multi-layer perceptron) — A fully connected feed-forward neural network, able to learn nonlinear relationships between input features and target.

Motif — A short recurring sequence pattern associated with a function. Mutating or removing motifs tests whether a model depends on them or has found signal beyond them.

PCA (principal component analysis) — A dimensionality-reduction technique that re-expresses many correlated features as a smaller number of uncorrelated components, ordered by how much of the total variance each captures. Plotting the first two components gives a two-dimensional view of a high-dimensional feature space.

Precision — Of the items predicted to be in a class, the proportion that truly are. Answers "when the model says activator, how often is it right?".

R^2 (coefficient of determination) — The share of variance in the target explained by the model. 1.0 is perfect; 0 is no better than predicting the mean.

Random Forest — An ensemble of decision trees, each trained on a bootstrap sample of the data with a random subset of features considered at each split; predictions are aggregated across trees.

Recall — Of the items truly in a class, the proportion the model finds. Answers "of all the real activators, how many did the model catch?".

Regularisation — A penalty on model complexity added to the training objective, to discourage overfitting. L2 (ridge) shrinks coefficients; L1 (lasso) can zero them.

Ridge regression — Linear regression with an L2 penalty on coefficient size — well suited to high-dimensional feature sets where predictors are correlated.

RMSE (root mean squared error) — The square root of the mean squared error. Like MAE but penalises large errors more heavily.

SHAP (SHapley Additive exPlanations) — A feature-attribution method that assigns each feature a signed contribution to each individual prediction, on a common scale. Unlike impurity-based importance it reveals direction as well as magnitude.

SHAP beeswarm plot — One row per feature, one dot per sequence. Colour encodes the feature value (red high, blue low); horizontal position is the signed SHAP contribution. Reading colour against position shows whether high values of a feature push predictions toward the positive or the negative class.

Sliding window — Computing features over short consecutive sub-sequences — here three residues wide, advancing one position at a time — so that local rather than whole-sequence properties are captured.

Spearman rank correlation — A correlation measured on ranks rather than raw values. High Spearman with low R^2 means the model orders items well even when its absolute predictions are off.

Support — The number of true instances of a class in the evaluation set — the denominator behind that class's recall.

Threshold optimisation — Choosing the cut-off applied to a model's continuous output to convert it into a class label, trading precision against recall.

Transfer learning — Reusing representations learned on a large general dataset (here, ESM pretrained on protein sequences) for a smaller specific task.

Univariate feature screening — Fitting a separate one-feature model for every feature and ranking features by their individual explanatory power (R^2), independent of any other feature.